Uppsala University

15 Credit Project

Topological Quantum Field Theory And Dimensional Reduction

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Abstract

Topological Quantum Field Theory (TQFT) provides a framework in which observables depend only on the topology of the underlying spacetime and are independent of the metric. In this paper, we review the foundational aspects of TQFT and classify such theories into Schwarz-type and Witten-type. The main contribution of this work is the dimensional reduction of twisted 4D $\mathcal{N} = 2$ Super Yang–Mills theory on the product manifold $\Sigma \times S^2$. By employing a spatially dependent "flipping projector," we derive an effective 2D action that matches precisely the $\mathcal{N} = (2, 2)$ equivariant sigma model constructed in two dimensions. This result demonstrates that the distinctive geometric constraints of the 2D theory arise directly from the geometry of the twisted 4D theory.

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1 Introduction

Topological quantum field theory (TQFT) is a type of quantum field theory (QFT) that computes topological invariants, specifically a QFT where certain correlation functions do not depend on the spacetime metric. Unlike traditional QFT where many physical quantities vary with energy scale and deal with the evolution and interaction of particles, TQFT emphasizes the description of how the topology of spacetime constrains/controls physical observables in the theory. The physical state space is not the entire Hilbert space, but rather the zero-energy quantum space of the Hamiltonian ($H = 0$), as will be explained later, where these states remain unchanged under the time evolution operator, exhibiting almost no dynamical evolution. As a consequence, the theory is independent of the background metric, allowing TQFT to typically avoid issues such as scale dependence, UV divergences, and renormalization. Its observables differ from examples like scattering cross-sections in QFT; instead, TQFT observables are topological invariants, such as links, knots, and Wilson loops. Moreover, certain QFT may reduce to a TQFT in some limits or energy scales; for example, at low energy limits or when a gap exists in condensed matter systems, topological behavior can emerge.

2 What is TQFT

To discuss TQFT, it is convenient to begin with a general QFT formulated on a flat spacetime background, such as Minkowski space. We adopt a right-handed coordinate system $x^\mu = (t, x^i)$, so that spacetime admits a $(3+1)$-dimensional decomposition. Topologically, this corresponds to a manifold of the form $M = \Sigma \times \mathbb{R}$, where Σ denotes the three-dimensional spatial slice and $\mathbb{R}$ represents the time direction.

Since physical interactions occur over a finite time interval, one may instead consider the spacetime manifold $M = \Sigma \times I$, where $I = [0, 1]$. In this case, M has two boundary components, $\Sigma_0 = \Sigma \times \{0\}$ and $\Sigma_1 = \Sigma \times \{1\}$. Compatibility with the orientation of spacetime implies that Σ_1 carries the opposite orientation, and the boundary may be written as $\partial M = \Sigma_0 \cup \Sigma_1^*$. The path integral on such a manifold defines a linear map between quantum states associated with the boundary components, that is, a map from the Hilbert space $Z(\Sigma)$ at Σ_0 to the Hilbert space $Z(\Sigma)$ at Σ_1. Formally, this may be interpreted as an evolution operator.

For a topological quantum field theory, however, the Hamiltonian is trivial, $H = 0$, so that both real and imaginary time evolution operators reduce to the identity, $e^{itH} = 1 \quad \text{and} \quad e^{-\tau H} = 1$. Consequently, propagation along the cylinder $\Sigma \times I$ is trivial: there is no physical time evolution, and the corresponding bordism acts as the identity on the Hilbert space $Z(\Sigma)$. In this sense, the theory has no real dynamics.

In Witten-type TQFTs, this absence of dynamics admits a natural cohomological interpretation. One introduces a nilpotent scalar operator Q satisfying $Q^2 = 0$, and defines physical observables as elements of the Q-cohomology. Since the energy–momentum tensor is Q-exact, $T_{\mu\nu} = \{Q, G_{\mu\nu}\}$, the Hamiltonian can likewise be written as a Q-exact operator, $H = \int_\Sigma T_{00} = \{Q, V\}$, and therefore acts trivially on physical states. As a result, correlation functions depend only on the cohomology classes of observables and encode purely topological information.

This viewpoint was first clearly formulated by Atiyah in 1988 [1], who provided an axiomatic definition of TQFT as a symmetric monoidal functor from the category of d-dimensional bordisms to finite-dimensional vector spaces. In this framework, closed $(d-1)$-dimensional manifolds are assigned state spaces, while d-dimensional bordisms are assigned linear maps between them.

Concretely, a d-dimensional TQFT is a symmetric monoidal functor

$$Z : \mathbf{Bord}_d \to \mathbf{Vect}.$$

Here, the objects of $\mathbf{Bord}_d$ are closed $(d-1)$-manifolds Σ, and the morphisms are bordisms

$$M : \Sigma_{\text{in}} \to \Sigma_{\text{out}}, \qquad \partial M = \Sigma_{\text{in}} \sqcup \Sigma_{\text{out}}.$$

Functoriality means

$$Z(M_1 \circ M_0) = Z(M_1) \circ Z(M_0), \qquad Z(\mathrm{id}_\Sigma) = \mathrm{id}_{Z(\Sigma)}.$$

The monoidal structure is encoded by

$$Z(\Sigma_0 \sqcup \Sigma_1) \cong Z(\Sigma_0) \otimes Z(\Sigma_1), \qquad Z(\varnothing) \cong \mathbb{C},$$

and symmetry requires invariance under exchange of disjoint components.

After Wick rotating the Minkowski spacetime, we obtain a theory defined on a Euclidean manifold with a metric $g_{\mu\nu}$. In this setting, the basic observables of QFT are usually correlation functions of local operators, which can be written in the path integral formulation as

$$\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_g = \frac{1}{Z[g]} \int \mathcal{D}\phi \; \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \; e^{-S[\phi;g]},$$

where the subscript g emphasizes the dependence on the background metric. The choice of metric affects not only the kinetic terms and the definition of operators, but also appears explicitly in the action through the Laplacian, covariant derivatives, and the measure. To understand this dependence, one usually considers infinitesimal variations of the metric, which introduces the definition of the energy–momentum tensor. By viewing the action as a functional of the metric, one can write

$$T^{\mu\nu}(x) = -\frac{2}{\sqrt{g(x)}} \frac{\delta S[\phi; g]}{\delta g_{\mu\nu}(x)},$$

which means that any change of the metric corresponds to an insertion of $T^{\mu\nu}$ in correlation functions:

$$\delta_g \langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle = -\frac{1}{2} \int d^d x \, \sqrt{g} \; \delta g_{\mu\nu}(x) \; \langle T^{\mu\nu}(x) \, \mathcal{O}_1 \cdots \mathcal{O}_n \rangle.$$

Thus, the energy–momentum tensor is the basic object describing the response of quantum field theory to the background geometry. Although if the theory is invariant under diffeomorphisms, the energy-momentum tensor satisfies the covariant conservation law

$$\nabla_\mu T^{\mu\nu} = 0,$$

this does not mean that the theory is independent of the metric. On the contrary, this conservation law is only a powerful physical constraint, which tells us that the physical laws of the theory do not depend on the choice of coordinates. However, any continuous deformation of the metric $\delta g_{\mu\nu}$ will cause a nonzero change in the observables of the theory, because this change is governed by the energy-momentum tensor itself. Therefore, ordinary quantum field theory must have nontrivial dependence on the metric.

Against this background, the special feature of TQFT becomes clear. Through topological twisting, the energy-momentum tensor itself becomes an exact "Q-exact" term, namely

$$T_{\mu\nu} = \{Q, G_{\mu\nu}\}.$$

According to the properties of supersymmetric theory, the expectation value of any Q-exact term is zero.

This leads to the expectation value of $T_{\mu\nu}$ being zero, and the derivative of the observable with respect to the metric becomes zero:

$$\frac{\delta}{\delta g_{\mu\nu}}\langle O\rangle = \langle\{Q,\ldots\}\rangle = 0.$$

so their values are topological invariants.

3 Classification of TQFTS

Algebraic Structures in Low-Dimensional TQFTs

Depending on the mechanism, two main types can be distinguished. The first is the so-called Schwarz type, such as three-dimensional Chern–Simons theory and certain BF models, whose actions themselves do not involve the metric and therefore directly yield metric-independent quantities. The second is the Witten type, usually arising from supersymmetric quantum field theories, where a topological twist produces a nilpotent operator Q. In these theories, variations of the metric can be written as Q-exact terms, and for all Q-closed operators the correlation functions are invariant under metric changes. Therefore, the physical observables in the Q-cohomology naturally define topological invariants.

Historically, Schwarz-type theories, such as three-dimensional Chern–Simons theory [2], produced topological invariants like the Jones polynomial of knots and three-manifold invariants. Witten-type theories, starting with twisted supersymmetric gauge theories, gave rise to Donaldson-Witten theory [4], which provides a physical method for computing Donaldson invariants of four-manifolds. Both kinds of TQFT have since influenced mathematics and physics, including applications in topological phases and anyonic excitations. In the axiomatic approach of Atiyah–Segal leads to concrete algebraic structures in low dimensions:

- **1D TQFT**: assigns to a point a vector space V over a field k. The cobordisms (intervals) correspond to linear maps between these vector spaces. Thus a 1D TQFT is essentially determined by a finite-dimensional k-vector space.

- **2D TQFT**: assigns to a circle an algebra A, more specifically a *Frobenius algebra*. A Frobenius algebra is a finite-dimensional k-algebra A together with a bilinear form

  $$\langle\cdot,\cdot\rangle : A\times A\to k$$

  which is *non-degenerate* and compatible with the multiplication:

  $$\langle ab, c\rangle = \langle a, bc\rangle.$$

  Here, "non-degenerate" means that if $\langle a,b\rangle = 0$ for all $b\in A$, then $a=0$. Equivalently, the matrix representation of $\langle\cdot,\cdot\rangle$ is invertible, i.e. full rank. Intuitively, this means that the bilinear form is sensitive enough to "detect" every element of the algebra: no nonzero element is invisible under the pairing.

  Geometrically, this bilinear form encodes the evaluation of closed surfaces (like the 2-sphere) to numbers, and the multiplication encodes the "pair-of-pants" cobordism (two circles merging into one). Thus, 2D TQFTs and Frobenius algebras are equivalent structures.

Since a 2D TQFT assigns a finite-dimensional vector space $A = Z(S^1)$ together with a Frobenius algebra structure. From a representation-theoretic point of view:

- The algebra A can be interpreted as an **algebra of observables**.
- The Frobenius pairing
$$\langle a, b\rangle = \varepsilon(ab)$$
defines a nondegenerate bilinear form, analogous to a trace or expectation value.
- The Frobenius condition implies an isomorphism
$$A \cong A^*$$
as A-modules.

Thus, a 2D TQFT may be viewed as a representation of an associative algebra equipped with a trace functional, where states and operators are unified in a single algebraic structure.

The correspondence between topology and algebra can be summarized as follows:

Topological Operation	Algebraic Structure
Circle S^1	Vector space A
Pair of pants (two inputs, one output)	Multiplication $\mu : A \otimes A \to A$
Pair of pants (one input, two outputs)	Comultiplication $\Delta : A \to A \otimes A$
Disk	Counit $\varepsilon : A \to k$

The equivalence of different decompositions of a surface translates into associativity and the Frobenius compatibility condition.

The Atiyah–Segal axioms describe a TQFT as a symmetric monoidal functor
$$Z : \mathrm{Bord}_2 \to \mathrm{Vect}_k.$$

This formulation admits a natural generalization to arbitrary symmetric monoidal categories $\mathcal{C}$. In this setting, the notion of a Frobenius algebra is replaced by that of a **Frobenius object**.

A **Frobenius object** in a symmetric monoidal category $\mathcal{C}$ is an object $A \in \mathcal{C}$ equipped with:

- a multiplication (monoid structure)
$$\mu : A \otimes A \to A, \quad \eta : \mathbf{1} \to A,$$
- a comultiplication (comonoid structure)
$$\Delta : A \to A \otimes A, \quad \varepsilon : A \to \mathbf{1},$$

such that:

- (A, μ, η) is an associative algebra object,
- (A, Δ, ε) is a coassociative coalgebra object,

- the Frobenius compatibility condition holds:

$$(\mathrm{id} \otimes \mu) \circ (\Delta \otimes \mathrm{id}) = \Delta \circ \mu = (\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \Delta).$$

This condition expresses the compatibility between multiplication and comultiplication, reflecting the topological invariance of the theory under different decompositions of cobordisms.

When $\mathcal{C} = \mathrm{Vect}_k$, this definition reduces to the usual notion of a Frobenius algebra. Thus, ordinary Frobenius algebras can be understood as special cases of Frobenius objects.

- **3D TQFT**: assigns to a circle a category. A prominent example is Chern–Simons theory, where to S^1 one assigns the modular tensor category of representations of a quantum group. Cobordisms correspond to functors between these categories, reflecting the fact that 3D topology naturally encodes categorical rather than purely algebraic data.

  A modular tensor category is a highly structured tensor category $\mathcal{C}$ with the following properties:

  - **Semisimplicity:** every object decomposes as a direct sum of simple objects.
  - **Finite simple objects:** there are finitely many isomorphism classes of simple objects, corresponding physically to distinct particle or line operator types.
  - **Monoidal (tensor) structure:** there is a bifunctor

$$\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C},$$

    describing the fusion of line operators.
  - **Braiding:** natural isomorphisms

$$c_{X,Y} : X \otimes Y \to Y \otimes X,$$

    encoding the topological exchange (braiding) of line operators.
  - **Duals:** every object has a dual, corresponding to antiparticles or orientation reversal.
  - **Non-degeneracy:** the braiding is non-degenerate, ensuring that the theory is fully topological (this is the modular condition).

These structures encode the essential topological data of the theory: fusion corresponds to tensor products, while braiding captures nontrivial linking and knotting phenomena in three dimensions.

Cobordisms as Functors

In the categorical setting, cobordisms no longer correspond merely to linear maps, but to functors between categories.

- A surface Σ with boundary S^1 defines a functor

$$Z(\Sigma) : \mathcal{C} \to \mathcal{C}.$$

- Gluing of cobordisms corresponds to composition of functors.

- More generally, one obtains a functorial assignment at the categorical level:

$$Z : \mathrm{Bord}_3 \to \mathrm{Cat},$$

where Cat denotes an appropriate category of tensor categories.

In this sense, 3D TQFTs provide a natural setting for *categorified representation theory*, where algebraic structures are replaced by higher-level categorical analogues.

- **4D TQFT**: even higher structures appear. To a surface one assigns a category, and to a circle one assigns a 2-category. Physically important examples include Donaldson–Witten theory and Seiberg–Witten theory. In these cases, the path integral localizes onto moduli spaces of gauge-theoretic objects (such as instantons or monopoles), and the resulting invariants (Donaldson invariants, Seiberg–Witten invariants) can be interpreted as topological data extracted from these moduli spaces. Conceptually, such 4D TQFTs can be viewed as producing "higher-categorical analogues" of Frobenius algebras, where the role of the non-degenerate bilinear form is played by integration over moduli spaces, which yields topological invariants of 4-manifolds.

4 Schwarz-Type TQFTs

A typical feature of Schwarz-type topological quantum field theories is that the action is written without the background metric. Therefore, at the classical level, its dynamical equations and many numerical quantities do not depend on the geometric metric of the manifold, but only on the smooth/oriented structure of the manifold and the connection between the gauge bundle.

Three dimensional Chern-Simons theory

Abelian Case

The three-dimensional Chern-Simons theory is a well-known example of a Schwarz-type TQFT. However, it is helpful to look at a wider picture. The main idea of topological invariance—that physical quantities depend only on the global shape or topology of space—appeared in simpler models even before Chern-Simons theory became popular in quantum field theory.

As an example, consider the Abelian case with gauge group $G = U(1)$. The Chern-Simons action is

$$S_{\mathrm{CS}}^{\mathrm{Abelian}}[A] = \frac{k}{4\pi} \int_M A \wedge dA, \tag{4.0.1}$$

which is clearly independent of the metric. This also defines a TQFT, but its topological features are simpler than those in the non-Abelian case. For example, the Abelian theory does not produce knot invariants like the Jones polynomial. Instead, its main topological observables are linking numbers of loops. The level k needs to be quantized to keep gauge invariance under large gauge transformations. Physically, this quantization arises from the compact nature of the gauge group $U(1)$. Unlike a non-compact group (like $\mathbb{R}$), the $U(1)$ group manifold is topologically a circle. This allows for large gauge transformations that wrap non-trivially around this circle (characterized by a winding number $n \in \mathbb{Z}$).

Quantization of the Level k in Abelian Theory

We can consider the theory on the manifold $M = S^1 \times S^2$. The gauge group is $U(1)$, which is topologically a circle. A gauge transformation is defined by a map $g : M \to U(1)$, acting on the connection as:

$$A \to A' = A + g^{-1}dg \tag{4.0.2}$$

Writing $g(x) = e^{i\theta(x)}$, this corresponds to $A \to A + id\theta$ (or $A + d\theta$ in real normalization). While gauge transformations (where g is continuously deformable to the identity) leave the action invariant, we must consider large gauge transformations that wind around the non-trivial cycles of the gauge group.

Consider a gauge transformation g that winds m times around the S^1 component of the spacetime M. Simultaneously, assume there is a magnetic monopole flux through the S^2 component. According to the Dirac quantization condition, the flux integral must be a multiple of 2π:

$$\int_{S^2} dA = 2\pi n, \quad n \in \mathbb{Z} \tag{4.0.3}$$

Under the gauge transformation $A' = A + g^{-1}dg$, the change in the action is:

$$\begin{aligned} \Delta S &= S[A'] - S[A] \\ &= \frac{k}{4\pi}\int_M (A + g^{-1}dg) \wedge d(A + g^{-1}dg) - \frac{k}{4\pi}\int_M A \wedge dA \\ &= \frac{k}{4\pi}\int_M (A \wedge d(g^{-1}dg) + g^{-1}dg \wedge dA + g^{-1}dg \wedge d(g^{-1}dg)) \end{aligned} \tag{4.0.4}$$

For an Abelian group, the term $g^{-1}dg \wedge d(g^{-1}dg)$ vanishes (as $d(g^{-1}dg) = d^2\theta = 0$). The term $A \wedge d(g^{-1}dg)$ also vanishes. We are left with the cross term:

$$\Delta S = \frac{k}{4\pi}\int_M g^{-1}dg \wedge dA \tag{4.0.5}$$

On the product manifold $M = S^1 \times S^2$, this integral factorizes:

$$\Delta S = \frac{k}{4\pi}\left(\oint_{S^1} g^{-1}dg\right)\left(\int_{S^2} dA\right) \tag{4.0.6}$$

Using the winding number m of the gauge transformation ($\oint_{S^1} g^{-1}dg = 2\pi i m$ or $2\pi m$ depending on normalization) and the magnetic flux n ($\int_{S^2} dA = 2\pi n$):

$$\Delta S = \frac{k}{4\pi}(2\pi m)(2\pi n) = \pi k m n \tag{4.0.7}$$

(Note: Depending on the precise normalization of the $U(1)$ generators, this is often written as $2\pi k m n$).

In the path integral formulation $Z = \int \mathcal{D}A\, e^{iS}$, the physics remains invariant only if the phase e^{iS} is single-valued. This requires the change in action ΔS to be an integer multiple of 2π:

$$\Delta S = 2\pi N, \quad N \in \mathbb{Z} \tag{4.0.8}$$

Substituting our result:

$$\pi k m n = 2\pi N \implies k = \frac{2N}{mn} \tag{4.0.9}$$

Since m and n can be any integers (e.g., take $m = 1, n = 1$), consistency implies that k must be an integer (specifically an even integer for purely bosonic manifolds, or any integer if the manifold admits a spin structure, which all 3-manifolds do).

Thus, gauge invariance under large gauge transformations enforces the quantization of the Chern-Simons level $k \in \mathbb{Z}$.

Non-Abelian case

For the non-Abelian case, we now proceed to show that this quantity is gauge invariant provided that k is integer. On a three-dimensional manifold M, the non-Abelian (with G a compact Lie group such as $SU(N)$) Chern–Simons action is given by

$$S_{\mathrm{CS}}[A] = \frac{k}{4\pi}\int_M \mathrm{Tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right), \tag{4.0.10}$$

where A is a Lie(G)-algebra valued connection one-form, and k is the coupling constant (known as the level). This expression is explicitly metric-independent and depends only on the connection and the topology of the manifold.
The partition function of the quantum theory is

$$Z = \int \mathcal{D}A\, e^{iS_{\mathrm{CS}}[A]}. \tag{4.0.11}$$

under the large gauge transformation:

$$A \to A^g = g^{-1}Ag + g^{-1}dg. \tag{4.0.12}$$

We defined:

$$B := g^{-1}dg, \qquad C := g^{-1}Ag. \qquad A^g = B + C$$

Using Maurer–Cartan equaton (a Lie-algebra valued form with vanishing curvature):

$$dB + B\wedge B = 0.$$

Derivative of C:

$$dC = g^{-1}dA\, g + C\wedge B - B\wedge C.$$

Trace cyclicity:

$$\mathrm{Tr}(X\wedge Y\wedge Z) = \mathrm{Tr}(Y\wedge Z\wedge X) = \mathrm{Tr}(Z\wedge X\wedge Y).$$

$$\begin{aligned} S_{\mathrm{CS}}[A^g] &= \mathrm{Tr}\big((C+B)\wedge d(C+B) + \tfrac{2}{3}(C+B)^3\big) \\ &= \mathrm{Tr}(C\wedge dC) + \mathrm{Tr}(B\wedge dC) + \mathrm{Tr}(C\wedge dB) + \mathrm{Tr}(B\wedge dB) \\ &\quad + \tfrac{2}{3}\mathrm{Tr}(C\wedge C\wedge C) + 2Tr(C\wedge C\wedge B) \\ &\qquad + 2Tr(C\wedge B\wedge B) + \tfrac{2}{3}\mathrm{Tr}(B\wedge B\wedge B). \end{aligned} \tag{4.0.13}$$

$$\mathrm{Tr}(C\wedge dC) = \mathrm{Tr}\big(C\wedge(g^{-1}dA\, g)\big) + \mathrm{Tr}(C\wedge C\wedge B) - \mathrm{Tr}(C\wedge B\wedge C), \tag{4.0.14}$$

$$\mathrm{Tr}(B\wedge dC) = \mathrm{Tr}\big(B\wedge(g^{-1}dA\, g)\big) + \mathrm{Tr}(B\wedge C\wedge B) - \mathrm{Tr}(B\wedge B\wedge C), \tag{4.0.15}$$

$$\mathrm{Tr}(C\wedge dB) = -\mathrm{Tr}(C\wedge B\wedge B), \tag{4.0.16}$$

$$\mathrm{Tr}(B\wedge dB) = -\mathrm{Tr}(B\wedge B\wedge B). \tag{4.0.17}$$

Combine the above into (4.0.13). First extract the pieces that assemble to

$$\begin{aligned} S_{\mathrm{CS}}[A^g] &= \mathrm{Tr}\Big(C\wedge(g^{-1}dAg)\Big) + \mathrm{Tr}\Big(B\wedge(g^{-1}dAg)\Big) \\ &\quad + \Big[\mathrm{Tr}(C\wedge C\wedge B) - \mathrm{Tr}(C\wedge B\wedge C) + \mathrm{Tr}(B\wedge C\wedge B) \\ &\qquad - \mathrm{Tr}(B\wedge B\wedge C) - \mathrm{Tr}(C\wedge B\wedge B) - \mathrm{Tr}(B\wedge B\wedge B)\Big] \\ &\quad + \tfrac{2}{3}\,\mathrm{Tr}(C^3) + 2\,\mathrm{Tr}(C^2B) + 2\,\mathrm{Tr}(CB^2) + \tfrac{2}{3}\,\mathrm{Tr}(B^3). \end{aligned} \tag{4.0.18}$$

where

$$\mathrm{Tr}(C \wedge (g^{-1} dA\, g)) = \mathrm{Tr}(A \wedge dA), \qquad \mathrm{Tr}(C \wedge C \wedge C) = \mathrm{Tr}(A \wedge A \wedge A),$$

because $C = g^{-1}Ag$ and Tr is invariant under conjugation and cyclic permutations. Hence

$$\mathrm{Tr}(C \wedge (g^{-1} dA\, g)) + \frac{2}{3}\,\mathrm{Tr}(C \wedge C \wedge C) = \mathrm{Tr}(A \wedge dA) + \frac{2}{3}\,\mathrm{Tr}(A \wedge A \wedge A) = S_{\mathrm{CS}}[A]. \quad (4.0.19)$$

$$\begin{aligned} S_{\mathrm{CS}}[A^g] - S_{\mathrm{CS}}[A] &= \mathrm{Tr}\,(B \wedge g^{-1} dA\, g) + \mathrm{Tr}(B \wedge B \wedge A) + \tfrac{1}{3}\,\mathrm{Tr}(B \wedge B \wedge B) \\ &= d\ \mathrm{Tr}(B \wedge A) + \tfrac{1}{3}\,\mathrm{Tr}(B \wedge B \wedge B). \\ &= \tfrac{1}{3}\,\mathrm{Tr}(B \wedge B \wedge B). \end{aligned}$$

With the usual normalization of the matrix trace

$$\mathrm{Tr}(T_a T_b) = \tfrac{1}{2}\,\delta_{ab},$$

the winding number of g is defined as [7]

$$n(g) \;=\; \frac{1}{24\pi^2} \int_M \mathrm{Tr}(g^{-1}dg)^3 \;\in\; \mathbb{Z}.$$

we get

$$S_{\mathrm{CS}}[A^g] - S_{\mathrm{CS}}[A] = \frac{k}{4\pi} \cdot \frac{1}{3} \int_M \mathrm{Tr}(g^{-1}dg)^3 = \frac{k}{4\pi} \cdot \frac{1}{3} \cdot (24\pi^2 n(g)) = 2\pi k\, n(g).$$

Under a gauge transformation $A \to A^g$, the action becomes $S_{\mathrm{CS}}[A] + 2\pi k n$. Therefore, in the path integral formulation, we require that $\exp(iS_{\mathrm{CS}})$ remains invariant under all gauge transformations. This is equivalent to demanding $e^{i2\pi k n(g)} = 1$ for any integer winding number $n(g)$ associated with the gauge transformation g. This leads to the quantization condition for the coupling constant: $k \in \mathbb{Z}$. This is the well-known physical explanation of "level quantization": if k is not an integer, the path integral becomes multi-valued under gauge transformations, breaking gauge invariance. As can be seen from this, the Chern–Simons theory, whose action is built from differential forms independent of the metric, has vanishing variation with respect to the metric, leading to an energy–momentum tensor $T_{\mu\nu} = 0$, and thus no local propagating degrees of freedom with physical observables depending only on the global topology of spacetime.

5 A Brief on Supersymmetry

To understand the construction of Witten-type TQFTs, a brief introduction to supersymmetry(SUSY) is essential. In this section, we briefly recall SUSY in four-dimensional (3+1) QFT, which provides the natural setting for the construction of Witten-type topological field theories. SUSY is a spacetime symmetry that extends the Poincaré symmetry by transforming bosonic states into fermionic states and vice versa. It postulates a relationship between particles of different spin statistics, In quantum mechanics, the SUSY transformation is provided by a supersymmetric operator Q as [8]

$$Q\,|\mathrm{Boson}\rangle \propto |\mathrm{Fermion}\rangle\,, \quad Q\,|\mathrm{Fermion}\rangle \propto |\mathrm{Boson}\rangle$$

In general, supersymmetry is generated by fermionic spinor operators known as supercharges, denoted by Q^A_α and $\bar{Q}_{A\dot{\alpha}}$. These operators extend the ordinary Poincaré algebra to

form the super-Poincaré algebra, where P_μ represents spacetime translations. Algebraically, the defining structure of supersymmetry is given by the following (anti-)commutation relations:

$$\{Q^A_\alpha, \bar{Q}_{B\dot{\beta}}\} = 2(\Gamma^\mu)_{\alpha\dot{\beta}} P_\mu \delta^A{}_B, \qquad \{Q^A_\alpha, Q^B_\beta\} = \{\bar{Q}_{A\dot{\alpha}}, \bar{Q}_{B\dot{\beta}}\} = 0,$$

$$[P_\mu, Q^A_\alpha] = [P_\mu, \bar{Q}_{A\dot{\alpha}}] = 0.$$

These relations express that the anticommutator of two supercharges produces a spacetime translation, implying that performing two successive supersymmetry transformations moves a field or state in spacetime. In Minkowski signature, one can consistently impose the Hermitian condition $(Q^A_\alpha)^\dagger = \bar{Q}_{A\dot{\alpha}}$, ensuring that the energy operator P^0 is real and positive. In Euclidean signature, however, Q and $\bar{Q}$ are treated as independent complex generators, and the R-symmetry acting on the supercharges is typically complexified rather than compact. The overall algebraic structure remains the same: supersymmetry unifies spacetime and internal symmetries through these spinorial generators.

Physically, supersymmetry implies that for every boson there should be a fermionic superpartner with the same mass and internal quantum numbers (and vice versa), forming irreducible representations called *supermultiplets*. For example, in four-dimensional $\mathcal{N} = 1$ supersymmetry, the chiral multiplet contains a complex scalar and a Weyl fermion, while the vector multiplet contains a gauge boson and its fermionic superpartner, the gaugino.

Although supersymmetry has not been observed experimentally at low energies, it remains a cornerstone of many theoretical frameworks beyond the Standard Model. If realized in nature, it must be a broken symmetry. Nevertheless, its mathematical consistency and rich structure make it an incredibly powerful tool for constructing quantum field theories with special properties, most notably for our purposes, the Witten-type TQFTs. The scalar supercharge Q central to these theories is obtained by a procedure called *topological twisting*, which redefines the Lorentz transformation properties of the supercharges using the theory's R-symmetry, as detailed in the following section.

Supersymmetric Quantum Mechanics

Supersymmetric quantum mechanics (SQM) is the cleanest setting for the cohomological idea used in Witten-type TQFTs. In SQM one has supercharges Q and $Q^\dagger$ such that

$$Q^2 = (Q^\dagger)^2 = 0, \qquad H = \frac{1}{2}\{Q, Q^\dagger\}. \tag{5.0.1}$$

States with $E > 0$ come in boson–fermion pairs related by Q. Unpaired states are supersymmetric ground states.

A standard realization identifies the Hilbert space with differential forms on a configuration space $\mathcal{M}$. In this identification, Q acts as a differential operator like the exterior derivative d. Thus the mathematical nilpotency

$$d^2 = 0 \tag{5.0.2}$$

is realized physically as

$$Q^2 = 0. \tag{5.0.3}$$

Supersymmetric ground states are then the Q-cohomology,

$$\mathcal{H}_{\text{SUSY}} = \frac{\ker Q}{\operatorname{im} Q}. \tag{5.0.4}$$

Morse theory (minimal definitions). Let $\mathcal{M}$ be a smooth manifold and $W : \mathcal{M} \to \mathbb{R}$ a smooth function. A point $p \in \mathcal{M}$ is a critical point if $dW(p) = 0$. In local coordinates $\{x^i\}$, the Hessian at p is

$$\mathrm{Hess}_p(W) = \left(\frac{\partial^2 W}{\partial x^i \partial x^j}(p) \right). \tag{5.0.5}$$

The critical point is non-degenerate if $\det(\mathrm{Hess}_p(W)) \neq 0$. A Morse function is a function whose critical points are all non-degenerate. The Morse index $\mu(p)$ is the number of negative eigenvalues of $\mathrm{Hess}_p(W)$.

Witten Deformation and Localization Given a Morse function W, we define the deformed differential (supercharge)

$$Q_t = e^{-tW} d e^{tW} = d + t dW \wedge, \quad Q_t^\dagger = e^{tW} d^\dagger e^{-tW}, \tag{5.0.6}$$

and the deformed Hamiltonian

$$H_t = \frac{1}{2}\{Q_t, Q_t^\dagger\} = \frac{1}{2}\left(-\Delta + t^2 |\nabla W|^2 + t\mathrm{Hess}(W)\right), \tag{5.0.7}$$

where the last term represents the fermionic coupling controlled by the Hessian of W.

For a large real number t, the potential term $t^2|\nabla W|^2$ dominates the Hamiltonian. Thus, the low-energy states concentrate near the critical points where $\nabla W = 0$.

Near a critical point p, we can approximate W by its quadratic term. This reduces the theory to a system of harmonic oscillators, with frequencies determined by $\mathrm{Hess}_p(W)$. This quadratic approximation is physically crucial for the path integral. In the localization limit (semi-classical approximation), the path integral becomes a **Gaussian integral** over fluctuations around the critical points. For this Gaussian integral to be well-defined, the Hessian matrix must have a non-zero determinant. This requires the critical points to be **non-degenerate**, which ensures they are isolated.

Furthermore, the Morse index $\mu(p)$ plays a central role in supersymmetry. Physically, supersymmetry guarantees that all high-energy states ($E > 0$) appear in **boson-fermion pairs**. Since they contribute with opposite signs, they cancel each other out completely in the partition function. **Only the zero-energy ground states (the critical points) can remain unpaired,** and thus the Witten index effectively counts the net number of these unpaired ground states. While the absolute contribution of each vacuum is normalized to 1. However, the **sign** of this contribution is non-trivial. It is determined by the number of negative eigenvalues of the Hessian. Specifically, each critical point p contributes a factor of $(-1)^{\mu(p)}$ to the topological invariant (the Witten index).

Thus, the localization mechanism sharpens the path integral onto the critical points, allowing us to compute the global topological invariant by simply summing the local signs $(-1)^{\mu(p)}$.

Link to Witten-type TQFT. In Witten's 4D construction the configuration space becomes infinite-dimensional (e.g. the space of gauge connections). The same cohomological idea survives: one keeps a nilpotent scalar charge Q with $Q^2 = 0$, and defines observables by Q-cohomology. Then adding Q-exact terms localizes the path integral to a finite-dimensional moduli space, which is the key step used in the next section.

6 Witten-type TQFTS

Unlike Schwarz type TQFT, Witten type TQFT is not such that its classical action is independent of the metric. On the contrary, they usually come from traditional supersymmetric quantum field theory (SQFT), these theories themselves contain local, propagating degrees of freedom, and their physical observables (such as scattering cross section) clearly depend on the spacetime metric. The cleverness of Witten type is that, through a procedure called topological twist, one can extract from such a theory that is sensitive to the metric a subset of observables, these observables are finally independent of the choice of the metric.

The core of this process is to construct a scalar supercharge, usually denoted as Q. In the twisted theory, Q becomes a nilpotent operator similar to the BRST charge, that is, gauge invariant operators under its action satisfy $Q^2 = 0$. This algebraic structure makes it possible to define Q-cohomology: the physical Hilbert space is constructed by Q-closed operators ($Q\mathcal{O} = 0$) modulo Q-exact operators ($\mathcal{O} = Q\{...\}$).

A basic property of supersymmetric theory is: the vacuum expectation value of any Q-exact operator is always zero. That is, for any operator G, one has

$$\langle\{Q, G\}\rangle = 0.$$

This is because Q is one of the generators of spacetime symmetry, and the vacuum is invariant under supersymmetry.

The topological property of Witten-type TQFT finally comes from a deeper fact: after twisting, the energy-momentum tensor $T_{\mu\nu}$ of the theory itself is a Q-exact term:

$$T_{\mu\nu} = \{Q, G_{\mu\nu}\},$$

where $G_{\mu\nu}$ is some fermionic operator. Now consider a Q-closed physical observable $\mathcal{O}$ (that is, $Q\mathcal{O} = 0$). The variation of its correlation function with respect to the metric can be written as:

$$\frac{\delta}{\delta g^{\mu\nu}}\langle\mathcal{O}\rangle = -\langle\mathcal{O}T_{\mu\nu}\rangle = -\langle\mathcal{O}\{Q, G_{\mu\nu}\}\rangle.$$

Since $\mathcal{O}$ is Q-closed, and Q is a conserved charge, we can move the action of Q to $\mathcal{O}$ (or in the path integral view, regard it as a symmetry transformation). Finally, using $Q^2 = 0$ and $Q|\text{vac}\rangle = 0$, we prove that the above variation is zero. Therefore, $\langle\mathcal{O}\rangle$ does not change under continuous deformation of the metric, and becomes a topological invariant. It means that all operators belonging to the Q-cohomology class (that is, all Q-closed gauge invariant operators), their correlation functions are topological invariants. These observables together form a "topological sector" in the original supersymmetric theory. Although the theory itself may still contain local degrees of freedom that depend on the metric, this special sector is only sensitive to the global topology of spacetime.

6.1 Construction of Witten type

Donaldson-Witten Theory

Donaldson invariants are smooth four-manifold invariants defined via the moduli space of anti-self-dual (ASD) connections ($F^+ = 0$) [3]. This section emphasizes the mechanism of tiwst. Supercharges are originally spinors, so they usually don't have covariantly constant solutions on curved manifolds. By diagonalizing the spin connection with a background R-symmetry connection — geometrically treating part of the R-symmetry bundle as part of the rotation

group — some supercharges can be turned into Lorentz scalars. These scalar supercharges can then exist globally on curved manifolds and lead to a BRST-like cohomological (topological) structure — this is called a **twist**.

After the mathematical definition of Donaldson invariants was rigorously established, Witten used QFT to reproduce and compute these invariants. We review Witten's construction [4] of four-dimensional TQFT. The key idea is to twist $\mathcal{N} = 2$ supersymmetric Yang–Mills theory, producing a scalar supercharge Q. The resulting theory is metric-independent, and its correlation functions reproduce Donaldson invariants of four-manifolds. We present the logical structure and essential mathematical steps of the construction.

6.1.1 The Action Before Twist

To construct the topological theory, Witten first formulates a supersymmetric Lagrangian in terms of the twisted field variables $(A_\alpha, \phi, \lambda, \eta, \psi_\alpha, \chi_{\alpha\beta})$. This initial Lagrangian, denoted as $\mathcal{L}_0$ in the original paper (Equation 2.7), represents the theory before the addition of the specific BRST-exact terms that facilitate localization. In flat Euclidean space, this is equivalent to the standard $\mathcal{N} = 2$ supersymmetric Yang-Mills action.

The explicit form of this action is:

$$\mathcal{L}_0 = \int d^4x\, Tr\left[\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} + \frac{1}{2}\phi D_\alpha D^\alpha \lambda - i\eta D_\alpha \psi^\alpha + iD_\alpha \psi_\beta \chi^{\alpha\beta} - \frac{i}{8}\phi[\chi_{\alpha\beta}, \chi^{\alpha\beta}] - \frac{i}{2}\lambda[\psi_\alpha, \psi^\alpha]\right] \tag{6.1.1}$$

This Lagrangian is invariant under the fermionic symmetry generated by the scalar supercharge Q

6.1.2 The Twist Mechanism: Mixing Spacetime and Internal Symmetries

The rotation group of Euclidean spacetime is $Spin(4)$, which is isomorphic to $SU(2)_L \times SU(2)_R$. The theory also possesses a global internal symmetry group $SU(2)_I \times U(1)_U$.

On a curved four-manifold, spinor fields are not always well-defined globally. To solve this, need to perform a Topological Twist.In the untwisted theory, fields are labeled by representations of the full symmetry group:

$$G_{total} = SU(2)_L \times SU(2)_R \times SU(2)_I \times U(1)_U \tag{6.1.2}$$

A representation is denoted by $(j_L, j_R, j_I)^q$, where j denotes the spin and q denotes the $U(1)_U$ charge.

The topological twist defines a new "twisted" right-handed rotation group, $SU(2)'_R$, which is the diagonal subgroup of the original spatial rotation $SU(2)_R$ and the internal symmetry $SU(2)_I$:

$$SU(2)'_R = \mathrm{diag}(SU(2)_R \times SU(2)_I) \tag{6.1.3}$$

Mathematically, this means the new right-handed spin j'_R is obtained by the tensor product of the original spins j_R and j_I. The decomposition follows the standard angular momentum addition rule of quantum mechanics:

$$j'_R = j_R \otimes j_I \longrightarrow |j_R - j_I| \oplus \cdots \oplus (j_R + j_I) \tag{6.1.4}$$

The left-handed spin j_L remains unchanged. We now apply this rule step-by-step to every field in the $\mathcal{N} = 2$ multiplet.

Decomposition of the Fields

(A) The Gauge Boson

- *Original Representation:* The gauge field A_α transforms as $(1/2, 1/2, 0)^0$.
- *Decomposition:*
  - $j_L = 1/2$ (Unchanged).
  - $j'_R = j_R \otimes j_I = 1/2 \otimes 0 = 1/2$.
- *Result:* The twisted field transforms as $(1/2, 1/2)^0$. This is the representation of a vector field (1-form) A_α.

(B) The Scalars

- *Original Representation:* The scalars transform as $(0, 0, 0)$ with charges $+2$ and -2.
- *Decomposition:*
  - $j_L = 0$.
  - $j'_R = 0 \otimes 0 = 0$.
- *Result:* They remain scalars $(0, 0)$. We denote them as ϕ (charge $+2$) and λ (charge -2).

(C) The Left-Handed Fermions

- *Original Representation:* These fermions transform as $(1/2, 0, 1/2)^1$.
- *Decomposition:*
  - $j_L = 1/2$.
  - $j'_R = j_R \otimes j_I = 0 \otimes 1/2 = 1/2$.
- *Result:* The resulting representation is $(1/2, 1/2)^1$. Geometrically, this is a vector field. We identify this with the 1-form fermion ψ_α.

(D) The Right-Handed Fermions

- *Original Representation:* These fermions transform as $(0, 1/2, 1/2)^{-1}$.
- *Decomposition:*
  - $j_L = 0$.
  - $j'_R = j_R \otimes j_I = 1/2 \otimes 1/2$.

  According to the addition of angular momenta, the product of two spin-1/2 states decomposes into a singlet (spin 0) and a triplet (spin 1):

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3} \quad \Longrightarrow \quad 1/2 \otimes 1/2 = 0 \oplus 1 \tag{6.1.5}$$

  Therefore, this fermion multiplet splits into two distinct geometric objects:

  1. A scalar component $(0, 0)^{-1}$. We identify this as the ghost field η.
  2. A tensor component $(0, 1)^{-1}$. We identify this as the self-dual 2-form $\chi_{\alpha\beta}$. (Note: $(0, 1)$ represents a self-dual tensor, while $(1, 0)$ would be anti-self-dual).

Summary of the Twisted Multiplet

The complete reorganization of the fields is summarized in Table 1.

Field	Original Rep. $(j_L, j_R, j_I)^q$	Twist Calculation $j'_R = j_R \otimes j_I$	Twisted Rep. $(j_L, j'_R)^q$	Geometric Object
A_α	$(1/2, 1/2, 0)^0$	$1/2 \otimes 0 = \mathbf{1/2}$	$(1/2, 1/2)^0$	Vector
ϕ, λ	$(0, 0, 0)^{\pm 2}$	$0 \otimes 0 = \mathbf{0}$	$(0, 0)^{\pm 2}$	Scalars
ψ_α	$(1/2, 0, 1/2)^1$	$0 \otimes 1/2 = \mathbf{1/2}$	$(1/2, 1/2)^1$	Vector (1-form)
η	$(0, 1/2, 1/2)^{-1}$	$1/2 \otimes 1/2 = \mathbf{0} \oplus \mathbf{1}$	$(0, 0)^{-1}$	Scalar (0-form)
$\chi_{\alpha\beta}$			$(0, 1)^{-1}$	Self-dual Tensor (2-form)

Table 1: Detailed decomposition of the $\mathcal{N} = 2$ multiplet under topological twisting. The transformation of the right-handed spin $j_R \to j'_R$ generates the integer-spin forms required for the topological theory.

6.1.3 The Twisted Action

Upon twisting, we replace $SU(2)_R$ with the diagonal sum of $SU(2)_R$ and $SU(2)_I$. The fields are reorganized into the multiplet $(A_\alpha, \phi, \lambda, \eta, \psi_\alpha, \chi_{\alpha\beta})$. On a general Riemannian four-manifold M with metric $g_{\alpha\beta}$, the twisted Lagrangian $\mathcal{L}$ is given by:

$$\begin{aligned}\mathcal{L} = \int_M d^4x\sqrt{g}\, Tr\Big[&\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} + \frac{1}{2}\phi D_\alpha D^\alpha \lambda - i\eta D_\alpha \psi^\alpha + iD_\alpha \psi_\beta \cdot \chi^{\alpha\beta} \\ &- \frac{i}{8}\phi[\chi_{\alpha\beta}, \chi^{\alpha\beta}] - \frac{i}{2}\lambda[\psi_\alpha, \psi^\alpha] - \frac{i}{2}\phi[\eta, \eta] - \frac{1}{8}[\phi, \lambda]^2\Big]\end{aligned} \tag{6.1.6}$$

This action is invariant under the nilpotent scalar supercharge Q, whose transformations are:

$$\begin{aligned}&QA_\alpha = i\psi_\alpha, \\ &Q\psi_\alpha = -D_\alpha \phi, \\ &Q\phi = 0, \\ &Q\lambda = 2i\eta, \\ &Q\eta = \tfrac{1}{2}[\phi, \lambda], \\ &Q\chi_{\alpha\beta} = F^+_{\alpha\beta} := F_{\alpha\beta} + \tilde{F}_{\alpha\beta}.\end{aligned}$$

One easily checks the closure up to gauge transformations, e.g.

$$Q^2 A_\mu = Q(i\psi_\mu) = i(-D_\mu \phi) = -iD_\mu \phi = \delta_\phi A_\mu,$$

so Q^2 acts as a gauge transformation with parameter ϕ.

Topological Invariance

To explicitly derive the relation $T_{\alpha\beta} = \{Q, \lambda_{\alpha\beta}\}$, one considers the variation of the action under an infinitesimal change in the metric $g_{\alpha\beta} \to g_{\alpha\beta} + \delta g_{\alpha\beta}$. The energy-momentum tensor is defined by:

$$\delta S = \frac{1}{2}\int_M \sqrt{g}\delta g^{\alpha\beta} T_{\alpha\beta}. \tag{6.1.7}$$

A subtle complication arises because the field $\chi_{\alpha\beta}$ is constrained to be a self-dual 2-form ($\chi_{\alpha\beta} = \frac{1}{2}\epsilon_{\alpha\beta\gamma\delta}\chi^{\gamma\delta}$). Since the definition of self-duality depends explicitly on the metric (via the volume

form and inverse metrics implicitly in the contraction), an arbitrary variation $\delta g_{\alpha\beta}$ would destroy this self-duality property if χ remained fixed.

To preserve the self-duality condition $\chi = *\chi$ under a metric deformation, $\chi_{\alpha\beta}$ must transform simultaneously. The required transformation is:

$$\delta\chi_{\alpha\beta} = \frac{1}{2}\epsilon_{\alpha\beta\gamma\delta}\delta g^{\gamma\sigma} g^{\delta\tau}\chi_{\sigma\tau} - \frac{1}{8}(\delta g^{\sigma\tau} g_{\sigma\tau})\chi_{\alpha\beta}. \tag{6.1.8}$$

This induced variation $\delta\chi_{\alpha\beta}$ contributes extra terms to the variation of the action δS. Taking this contribution into account is crucial for obtaining the correct expression for $T_{\alpha\beta}$.

After a somewhat lengthy computation incorporating these variations, one finds that the resulting energy-momentum tensor can indeed be written as a Q-exact:

$$T_{\alpha\beta} = \{Q, \lambda_{\alpha\beta}\} \tag{6.1.9}$$

where $\lambda_{\alpha\beta}$ is explicitly given by:

$$\begin{aligned}\lambda_{\alpha\beta} = &\tfrac{1}{2}\operatorname{Tr}\left(F_{\alpha\sigma}\chi_\beta{}^\sigma + F_{\beta\sigma}\chi_\alpha{}^\sigma - \tfrac{1}{2}g_{\alpha\beta}F_{\sigma\tau}\chi^{\sigma\tau}\right)\\ &+ \tfrac{1}{2}\operatorname{Tr}\left(\psi_\alpha D_\beta\lambda + \psi_\beta D_\alpha\lambda - g_{\alpha\beta}\psi_\sigma D^\sigma\lambda\right)\\ &+ \tfrac{1}{4}g_{\alpha\beta}\operatorname{Tr}(\eta[\phi,\lambda]).\end{aligned} \tag{6.1.10}$$

For any Q-closed operator $\mathcal{O}$, the correlation function $\langle\mathcal{O}\rangle$ is therefore metric-independent:

$$\frac{\delta}{\delta g^{\alpha\beta}}\langle\mathcal{O}\rangle = -\langle\mathcal{O}T_{\alpha\beta}\rangle = -\langle\mathcal{O}\{Q,\lambda_{\alpha\beta}\}\rangle = 0.$$

The Q-cohomology of operators thus defines the topological observables. In Donaldson-Witten theory, their correlation functions compute the Donaldson invariants of the 4-manifold M.

When considering the quantum theory, the partition function includes the coupling constant e and a topological term :

$$Z = \int(\mathcal{D}X)\exp\left(-\frac{1}{e^2}\mathcal{L}'\right) \tag{6.1.11}$$

where $\mathcal{L}'$ includes the topological term $\frac{1}{4}\int_M\sqrt{g}TrF_{\alpha\beta}\tilde{F}^{\alpha\beta}$ (which is proportional to the instanton number $\int TrF\wedge F$). The crucial feature of this action is that the dynamical part (scaled by $1/e^2$) is Q-exact. Specifically, $\mathcal{L}' = \{Q, V\}$ for a specific functional V.

Behavior of the Coupling

The topological nature of the theory is intimately linked to the coupling constant e. The variation of the partition function with respect to the coupling is:

$$\frac{\partial Z}{\partial e^2} = \frac{\partial}{\partial e^2}\int(\mathcal{D}X)\exp\left(-\frac{1}{e^2}\{Q,V\}\right) = \frac{1}{e^4}\int(\mathcal{D}X)\{Q,V\}\exp\left(-\frac{1}{e^2}\mathcal{L}'\right) = \frac{1}{e^4}\langle\{Q,V\}\rangle \tag{6.1.12}$$

Since the vacuum expectation value of a Q-exact operator vanishes ($\langle\{Q,\dots\}\rangle = 0$) in a supersymmetric theory, the theory is independent of the coupling constant e (as long as $e \neq 0$). This metric independence allows us to compute invariants in the weak coupling limit $e \to 0$, where the path integral localizes to the instantons.

6.2 Localization

The concept of localization in this 4D theory can be understood intuitively by looking at the simpler 1D SQM. In our four-dimensional case, the situation is an infinite-dimensional generalization of this idea. In fact, this is exactly the perspective of Floer theory, which is essentially an infinite-dimensional version of Morse theory. Here, the Morse function is replaced by the Yang-Mills action (or the Chern-Simons functional), and the critical points correspond to the solutions of the BPS equations, namely the instantons ($F^+ = 0$). Therefore, just as the 1D theory counts critical points, the path integral in the 4D twisted theory localizes onto the moduli space of instantons (defined modulo gauge equivalence). This procedure is a concrete manifestation of the Q-exact structure and its relation to topological invariance.

Consider the partition function with parameter t (which corresponding to $1/e^2$ in witten's paper):

$$Z(t) = \int [\mathcal{D}\Phi]\, e^{-t\{Q,V\} - S_{\text{top}}}.$$

Due to the cohomological nature of the path integral with respect to Q, the value of $Z(t)$ (and the expectation value $\langle \mathcal{O} \rangle$ of any Q-closed observable $\mathcal{O}$) is independent of the parameter t. To ensure the validity of the localization argument, the functional V must be chosen such that the bosonic part of the deformation term is positive semi-definite:

$$\{Q, V\}_{\text{bos}} \geq 0.$$

This positivity condition is crucial. It guarantees that as we take the limit $t \to \infty$, the factor $e^{-t\{Q,V\}_{\text{bos}}}$ acts as a damping factor, suppressing contributions from field configurations where $\{Q, V\}_{\text{bos}} > 0$. The path integral therefore receives contributions only from the configurations where the deformation term vanishes:

$$\{Q, V\}_{\text{bos}} = 0.$$

The configurations satisfying $\{Q, V\}_{\text{bos}} = 0$ are precisely the fixed points of the supersymmetry transformation generated by Q (typically, $\delta\Psi = 0$ where Ψ are the fermionic partners of the bosonic fields under Q). These equations are generally known as the BPS equations specific to the charge Q and the chosen functional V. A well-known example arises in the context of topologically twisted $\mathcal{N} = 2$ supersymmetric Yang-Mills theory on a four-manifold (Donaldson–Witten theory). With a standard choice for Q and V, the resulting BPS equations reduce precisely to:

$$F^+ = 0,$$

i.e., the ASD connection condition. In this specific case, the path integral localizes to the instanton moduli space.

The correlation functions of descent operators correspond precisely to Donaldson invariants. Insertions map to cohomology classes on the ASD moduli space, and the path integral computes their intersection numbers. Thus Witten's TQFT provides a physical realization of Donaldson theory.

6.3 Example: A-Twist of the 2D $\mathcal{N} = (2,2)$ Nonlinear σ-Model and the Case on S^2

The two-dimensional $\mathcal{N} = (2,2)$ supersymmetric nonlinear sigma model admits topological twisting. The A-twist can be defined for a Kähler target space, while the B-twist requires the target space to be Calabi–Yau.

The A-twist results in the topological A-model, whose correlation functions are related to the Gromov-Witten invariants of the target space, which count holomorphic curves. The BRST exactness of the action ensures metric independence of correlation functions. On the sphere S^2, the path integral localizes to the moduli space of holomorphic maps, yielding explicit results for the partition function and correlation functions.

Let Σ be a 2D Riemannian worldsheet with metric $h_{\alpha\beta}$, and let X be a compact Kähler manifold of complex dimension n with metric $g_{i\bar{j}}$ and Kähler form $\omega = ig_{i\bar{j}}dz^i \wedge d\bar{z}^{\bar{j}}$.Then we consider fermion fields: $\psi^i_+, \psi^{\bar{i}}_+, \psi^i_-, \psi^{\bar{i}}_-$ (spinors on Σ valued in pullbacks of TX) with $\psi^i_\pm \in \Gamma\Big(K^{1/2}_\pm \otimes \phi^* T^{1,0}X\Big)$, $\bar{\psi}^i_\pm \in \Gamma\Big(K^{1/2}_\pm \otimes \phi^* T^{0,1}X\Big)$ and bosonic fields: $\phi^i, \phi^{\bar{i}}$ (coordinates in X). The theory has $\mathcal{N} = (2,2)$ supersymmetry with supercharges $Q_+, \bar{Q}_+, Q_-, \bar{Q}_-$.

The action can be written as:

$$S = \int_\Sigma d^2\sigma\sqrt{h}\Big[g_{i\bar{j}}h^{\alpha\beta}\partial_\alpha\phi^i\partial_\beta\phi^{\bar{j}} + ig_{i\bar{j}}\psi^{\bar{j}}_- D_z\psi^i_+ + ig_{i\bar{j}}\psi^{\bar{j}}_+ D_{\bar{z}}\psi^i_- \\ + R_{i\bar{j}k\bar{l}}(\psi^i_+\psi^{\bar{j}}_+\psi^k_-\psi^{\bar{l}}_-)\Big], \tag{3.1}$$

with

$$D_z\psi^i_- = \partial_z\psi^i_- + \Gamma^i_{jk}\,\partial_z\phi^j\psi^k_-. \tag{3.2}$$

where $R_{i\bar{j}k\bar{l}}$ is the Riemann tensor of X.

In a typical QFT, the spin of a field is determined by its transformation under the Lorentz group $SO(2) \cong U(1)_L$. The A twist redefines the Lorentz group by mixing it with the vector R-symmetry $U(1)_V$, which makes some supercharges scalar: $U(1)'_L = \mathrm{diag}(U(1)_L \times U(1)_V)$. Under this twist, the fields transform with new spin assignments:

Field	Original Spin	Twisted Spin	Ghost Number
$\phi^i, \phi^{\bar{i}}$	0	0	0
ψ^i_+	$+\frac{1}{2}$	0	+1
$\psi^{\bar{i}}_+$	$+\frac{1}{2}$	+1	-1
ψ^i_-	$-\frac{1}{2}$	+1	-1
$\psi^{\bar{i}}_-$	$-\frac{1}{2}$	0	+1

We redefine the fermionic fields [5]:

$$\begin{aligned} \chi^i &= \psi^i_+ \quad \text{(scalar, ghost number +1)} \\ \chi^{\bar{i}} &= \psi^{\bar{i}}_- \quad \text{(scalar, ghost number +1)} \\ \rho^{\bar{i}}_z &= \psi^{\bar{i}}_+ \quad \text{(vector, ghost number -1)} \\ \rho^i_{\bar{z}} &= \psi^i_- \quad \text{(vector, ghost number -1)} \end{aligned}$$

Twisted fermions:

$\chi^i, \bar{\chi}^{\bar{i}}$ are worldsheet scalar fermions, $\rho^i_z\,dz$ is a $(1,0)$-form, $\rho^{\bar{i}}_{\bar{z}}\,d\bar{z}$ is a $(0,1)$-form, all taking values in the appropriate pullback bundles.

BRST Transformations

The scalar supercharge $Q_A = \bar{Q}_+ + Q_-$ becomes the BRST operator. Its action defines the

BRST transformations:

$$\begin{aligned}
Q_A \cdot \phi^i &= \chi^i, \\
Q_A \cdot \phi^{\bar{i}} &= \chi^{\bar{i}}, \\
Q_A \cdot \chi^i &= 0, \\
Q_A \cdot \chi^{\bar{i}} &= 0, \\
Q_A \cdot \rho_z^{\bar{i}} &= 2\partial_z \phi^{\bar{i}} - \chi^{\bar{j}} \Gamma^{\bar{i}}_{\bar{j}\bar{k}} \rho_z^{\bar{k}}, \\
Q_A \cdot \rho_{\bar{z}}^i &= 2\partial_{\bar{z}} \phi^i - \chi^j \Gamma^i_{jk} \rho_{\bar{z}}^k,
\end{aligned}$$

where Γ^i_{jk} are the Christoffel symbols of $g_{i\bar{j}}$.These transformations satisfy $Q_A^2 = 0$ on-shell.

Twisted Action
The twisted action can be written as:

$$S = \{Q_A, V\} + S_{\text{top}}, \tag{3.3}$$

where the topological term is:

$$S_{\text{top}} = -i \int_\Sigma \phi^* \omega = -i \int_\Sigma d^2 z \, \omega_{i\bar{j}} (\partial_z \phi^i \partial_{\bar{z}} \phi^{\bar{j}} - \partial_{\bar{z}} \phi^i \partial_z \phi^{\bar{j}}), \tag{3.4}$$

It depends only on the homotopy class of the map ϕ because it evaluates the pullback of the closed form ω on Σ. For a closed worldsheet, $\int_\Sigma \phi^* \omega = 2\pi d$, where d $\in \mathbb{Z}$ is the degree of the map. and we choose:

$$V = \int_\Sigma d^2 z \, g_{i\bar{j}} \left(\rho_z^{\bar{j}} \partial_{\bar{z}} \phi^i + \rho_{\bar{z}}^i \partial_z \phi^{\bar{j}} \right). \tag{3.5}$$

Explicit computation gives:

$$\begin{aligned}
\{Q_A, V\} = \int_\Sigma d^2 z \Big[& g_{i\bar{j}} (\partial_z \phi^i \partial_{\bar{z}} \phi^{\bar{j}} + \partial_{\bar{z}} \phi^i \partial_z \phi^{\bar{j}}) \\
& + i g_{i\bar{j}} \chi^{\bar{j}} \nabla_z \rho_{\bar{z}}^i + i g_{i\bar{j}} \chi^i \nabla_{\bar{z}} \rho_z^{\bar{j}} \\
& - R_{i\bar{j}k\bar{l}} \rho_z^{\bar{j}} \rho_{\bar{z}}^i \chi^k \chi^{\bar{l}} \Big].
\end{aligned} \tag{3.6}$$

This is exactly the original action minus the topological term.

Observables and Topological Invariance
Physical observables are Q_A-cohomology classes. For closed forms $\alpha_{(p,q)} \in \Omega^{p,q}(X)$, we construct:

$$\mathcal{O}_\alpha = \alpha_{i_1 \cdots i_p \bar{j}_1 \cdots \bar{j}_q} \chi^{i_1} \cdots \chi^{i_p} \chi^{\bar{j}_1} \cdots \chi^{\bar{j}_q}.$$

These satisfy $Q_A \cdot \mathcal{O}_\alpha = 0$ and correspond to cohomology classes in $H^{p,q}(X)$.
The energy-momentum tensor is Q_A-exact:

$$T_{\alpha\beta} = \frac{\delta S}{\delta h^{\alpha\beta}} = \{Q_A, b_{\alpha\beta}\}, \tag{3.7}$$

where $b_{\alpha\beta} = \frac{\delta V}{\delta h^{\alpha\beta}}$. This implies that correlation functions of Q_A-closed operators are metric-independent:

$$\frac{\delta}{\delta h^{\alpha\beta}} \langle \prod_a \mathcal{O}_a \rangle = -\langle T_{\alpha\beta} \prod_a \mathcal{O}_a \rangle = -\langle \{Q_A, b_{\alpha\beta}\} \prod_a \mathcal{O}_a \rangle = 0. \tag{3.8}$$

Therefore, the correlator is independent of the metric $h_{\alpha\beta}$.

Localization on S^2
Let the worldsheet be S^2 with metric:

$$ds^2 = \frac{4R^2}{(1+|z|^2)^2} dz d\bar{z}, \tag{3.9}$$

where z is a complex coordinate via stereographic projection and R is the radius.
The topological term becomes:

$$S_{\text{top}} = -i \int_{S^2} \phi^* \omega = -2\pi i d, \tag{3.10}$$

where $d = \frac{1}{2\pi} \int_{S^2} \phi^* \omega$ is the degree of the map $\phi : S^2 \to X$.

Q_A-Invariant Configurations and Localization
The path integral localizes to Q_A-invariant field configurations:

$$\begin{aligned} Q_A \cdot \phi^i &= 0 \Rightarrow \chi^i = 0, \\ Q_A \cdot \phi^{\bar{i}} &= 0 \Rightarrow \chi^{\bar{i}} = 0, \\ Q_A \cdot \rho_z^{\bar{i}} &= 0 \Rightarrow \partial_z \phi^{\bar{i}} = 0, \\ Q_A \cdot \rho_{\bar{z}}^{i} &= 0 \Rightarrow \partial_{\bar{z}} \phi^{i} = 0. \end{aligned} \tag{3.11}$$

These equations imply: The fermions χ are zero and the map ϕ is holomorphic: $\bar{\partial}\phi^i = 0$ and $\partial\phi^{\bar{i}} = 0$.

Thus, the infinite-dimensional path integral reduces to a finite-dimensional integral over the moduli space of holomorphic maps:

$$\mathcal{M} = \bigcup_{d \geq 0} \mathcal{M}_d, \quad \mathcal{M}_d = \{\phi : S^2 \to X \mid \bar{\partial}\phi = 0, \deg \phi = d\}. \tag{3.12}$$

The partition function becomes:

$$Z(S^2) = \int \mathcal{D}\phi \mathcal{D}\chi \mathcal{D}\rho \, e^{-S} = \sum_{d \geq 0} e^{2\pi i d t} Z_d, \tag{3.13}$$

where t is the complexified Kähler parameter and Z_d is the integral over $\mathcal{M}_d$.

Example: $X = \mathbb{CP}^1$
For $X = \mathbb{CP}^1$, degree d holomorphic maps $\phi : S^2 \to \mathbb{CP}^1$ are given by rational functions:

$$\phi(z) = \frac{P_d(z)}{Q_d(z)}, \tag{3.14}$$

where P_d, Q_d are degree d polynomials. The moduli space $\mathcal{M}_d$ has dimension $2d+1$.
The partition function is:

$$Z(S^2) = \sum_{d \geq 0} e^{2\pi i d t} \chi(\mathcal{M}_d) = \sum_{d \geq 0} e^{2\pi i d t} = \frac{1}{1 - e^{2\pi i t}}, \tag{3.15}$$

where $\chi(\mathcal{M}_d) = 1$ is the Euler characteristic of $\mathcal{M}_d$.

Consider the observable $\mathcal{O}_H$ corresponding to the Kähler form ω. The three-point function is:

$$\langle \mathcal{O}_H \mathcal{O}_H \mathcal{O}_H \rangle = \sum_{d \geq 0} e^{2\pi i d t} \int_{\mathcal{M}_d} \omega \wedge \omega \wedge \omega. \tag{3.16}$$

Since $\dim \mathcal{M}_d = 2d + 1$ and $\omega \in H^2(\mathcal{M}_d)$, the integral is non-zero only for $d = 0$:

$$\langle \mathcal{O}_H \mathcal{O}_H \mathcal{O}_H \rangle = 1. \tag{3.17}$$

We have constructed the topological A-model from the N=(2,2) nonlinear sigma model through the A-twist procedure. The BRST exactness of the action ensures metric independence of correlation functions. On S^2, the path integral localizes to the moduli space of holomorphic maps, yielding explicit results for the partition function and correlation functions. All results depend only on the degree d and the topology of the moduli space, and are independent of the metric of S^2 (such as the radius R).

7 Dimensional Reduction of 4D Twisted SYM on $\Sigma \times S^2$ via the Flipping Projector

A basic idea in TQFT is that theories in different dimensions can be related to each other. If a theory is topological, its observables do not depend on the metric. As a result, when the spacetime is a product manifold, one can shrink one factor without changing the theory, and obtain an equivalent description in lower dimension.

The goal of this work is to show that a similar relation exists even when the theory is not strictly topological. In the second part of this thesis, we study theories that are *almost* topological. These theories are insensitive to many deformations of the metric, but they can still depend on additional geometric data, such as the presence of a symmetry generated by a Killing vector field.

This situation naturally leads to the idea of an *equivariant* topological twist. In this case, the scalar supercharge does not square to zero. Instead, it closes on a symmetry of the background geometry. Schematically, one has

$$Q^2 \sim \mathcal{L}_v, \tag{7.0.1}$$

where $\mathcal{L}_v$ is the Lie derivative along a vector field v that generates an isometry of the spacetime.

From this point of view, our dimensional reduction relates a four-dimensional equivariantly twisted gauge theory to an effective two-dimensional model. This explains how the special geometric structure of the two-dimensional theory arises from the geometry and symmetry of the four-dimensional theory. It also places our construction in the same general framework as other equivariant twists that appear in localization and instanton partition functions.

In this section, our reduction follows the similar compactification strategy as developed in the work by Vafa [10], where shrinking one component of the product manifold localizes the path integral onto the moduli space of flat connections, we consider the twisted $\mathcal{N} = 2$ Super Yang-Mills theory on the product manifold $M^4 = \Sigma \times S^2$, where Σ is a Riemann surface and S^2 possesses a rotational $U(1)$ symmetry generated by a killing vector v. The reduction is performed by shrinking the Riemann surface Σ to zero size (limit $\epsilon_2 \to 0$). In this limit, we only keep the zero modes of the fields along Σ.

First, we need to introduce the new mode here. Standard Donaldson-Witten theory is too restrictive because it relies on a global self-duality condition (requiring $F^+ = 0$ everywhere). However, Pestun's [11] localization result on the sphere S^4 reveals a different structure: it includes contributions from instantons at the North Pole and anti-instantons at the South Pole.

The standard theory cannot describe this mixture of opposite topological sectors on a single manifold. To solve this, Festuccia et al [12] introduced the 'flipped projector' P^+_ω. Unlike the standard projector, this operator is spatially dependent. It behaves as a self-dual operator near "+" fixed points and smoothly flips to an anti-self-dual operator near "−" fixed points. This mechanism allows the theory to effectively glue the holomorphic and anti-holomorphic sectors together using the killing vector field v, the theory becomes transversely elliptic with respect to the $U(1)$ action generated by v. This means the ellipticity holds in the directions normal to the vector field orbits, requiring us to work within the framework of equivariant cohomology. The key to this gluing is a reflection map m that acts as a transition function between the two sectors. Given the dual 1-form $\kappa = g(v)$, the map

$$m : \Omega^{2+} \to \Omega^{2-} \tag{7.0.2}$$

is defined by

$$m(B) = -B + \frac{2}{\iota_v \kappa}\, \kappa \wedge \iota_v B. \tag{7.0.3}$$

Using this map, the flipping projector P^+_ω can be written explicitly as an interpolation determined by a function $h = \cos\omega$ (where $h = 1$ at the North Pole and $h = -1$ at the South Pole):

$$P^+_\omega = \frac{1}{1+h^2}(1 + h\, *_4 - s^2 m), \qquad s = \sin\omega. \tag{7.0.4}$$

This operator projects onto the new bundle $P^+_\omega \Omega^2$, smoothly connecting the holomorphic and anti-holomorphic sectors across the manifold.

Unlike standard dimensional reductions, we perform an equivariant reduction based on the localization principle, effectively projecting the theory onto the cohomology of the rotational symmetry ($\mathcal{L}_v \Phi = 0$, where Φ denotes the collective fields $\{A, \Psi, \chi, \dots\}$). By explicitly decomposing the 4D fields and integrating out the modes that do not satisfy the mixed-index projection constraints, we derive the complete 2D effective theory. This procedure yields an equivariant non-linear sigma model on Σ.

By explicitly performing the reduction, we derive the effective 2D action and the specific 'flipped' projector on S^2. Crucially, our results precisely match the intrinsically constructed 2D $\mathcal{N} = (2,2)$ theory proposed in [13]. This provides a derivation of the 2D equivariant sigma model and demonstrates that its novel self-duality condition is the direct dimensional descendant of the 4D twisted theory.

The 4D theory is defined by the "Twisting with a Flip" formalism [12]. The relevant 4D fields obey the cohomological transformations:

$$\begin{aligned} \delta A &= i\Psi, \quad \delta\Psi = \iota_v F + i d_A \phi, \\ \delta\chi &= H, \quad \delta H = i\mathcal{L}_v \chi - i[\phi, \chi]. \end{aligned} \tag{7.0.5}$$

7.1 Dimensional Reduction of Forms on $M^4 = \Sigma \times S^2$

Elliptic operators in localization

In this section we explain one basic mathematical point. It is important for the localization method and for the dimensional reduction. The point is that the kinetic operators coming from the Q-exact term are elliptic operators on the compact manifold $M_4 = \Sigma \times S^2$.

Definition. Let P be a differential operator of order m

$$P : \Gamma(E) \longrightarrow \Gamma(F)$$

between sections of vector bundles E and F over M_4. The highest-derivative part of P defines the principal symbol

$$\sigma_m(P)(x,\xi),$$

which is a homogeneous polynomial of degree m in the cotangent vector $\xi \neq 0$. The operator is called *elliptic* if $\sigma_m(P)(x,\xi)$ is invertible for all $x \in M_4$ and all $\xi \neq 0$.

Examples. The Laplace operator

$$\Delta = -\nabla^\mu \nabla_\mu$$

is elliptic, because $\sigma_2(\Delta)(\xi) = |\xi|^2 > 0$ for $\xi \neq 0$. A Dirac-type operator has principal symbol $\gamma^\mu \xi_\mu$, which is invertible when $\xi \neq 0$. Therefore the square of a Dirac operator is elliptic. In twisted $N = 2$ supersymmetric gauge theory, the operators K_Φ from the Q-exact kinetic term are of these types.

Why ellipticity is important. Since $M_4 = \Sigma \times S^2$ is compact and without boundary, ellipticity implies three key properties:

1. The spectrum is discrete.
2. The space of zero modes, $\ker K$, is finite-dimensional.
3. If $K\Phi = f$ and f is smooth, then Φ is also smooth (elliptic regularity).

These facts allow us to expand any field as

$$\Phi = \sum_n a_n v_n, \quad K v_n = \lambda_n v_n.$$

Non-zero modes ($\lambda_n \neq 0$) have a positive quadratic term in the action and are integrated out by Gaussian integrals. Their effect is the one-loop determinant $\det{}' K$. Zero modes ($\lambda_n = 0$) do not appear in the quadratic term and remain as finite-dimensional degrees of freedom.

Remark on zero modes and localization. In the original 4D twisted theory the localization deformation is of the form

$$S_{\text{loc}} = t\, QV,$$

where V is linear in fermionic fields. Therefore the deformation is quadratic in the fluctuations around the localization locus. It can always be written in the schematic form

$$S^{(2)}_{\text{loc}} = \langle \Phi, K\Phi \rangle,$$

with Φ collecting all bosonic and fermionic fields, and K a linear differential operator of Laplace or Dirac type. We do not need the explicit form of K here; only its ellipticity on the compact manifold $M_4 = \Sigma \times S^2$ is essential for the dimensional reduction.

Non-zero modes ($\lambda_n \neq 0$) appear with a positive quadratic term in the action. They are integrated out by Gaussian integrals and contribute only a one-loop determinant. Zero modes ($\lambda_n = 0$) do not appear in the quadratic term. They remain as finite-dimensional degrees of freedom.

Remark on boundary conditions. We assume that Σ and S^2 are compact without boundary. Thus M_4 has no physical boundary. The elliptic operators then have natural self-adjoint extensions, and integrations by parts do not give boundary terms. The case with a physical boundary requires supersymmetric and gauge-invariant boundary conditions. We do not study this case here.

1. Gauge Field A

We decompose the 4D 1-form gauge field $A \in \Omega^1(M^4)$ into components along Σ and S^2:

$$A(x,y) = A_\Sigma + A_{S^2} = A_a(x,y)dx^a + A_i(x,y)dy^i. \tag{7.1.1}$$

Here, x denotes coordinates on Σ, and y denotes coordinates on S^2. Shrinking Σ implies that the coefficients A_a and A_i do not depend on x.

The curvature $F = dA + A \wedge A$ decomposes according to the form degree on (Σ, S^2) as:

$$F = F_{\Sigma\Sigma} + F_{\Sigma S} + F_{SS}, \tag{7.1.2}$$

where:

- $F_{\Sigma\Sigma} = F_{ab}(x,y)dx^a \wedge dx^b$ is a (2,0)-form.
- $F_{\Sigma S} = F_{ai}(x,y)dx^a \wedge dy^i$ is a (1,1)-form.
- $F_{SS} = F_{ij}(x,y)dy^i \wedge dy^j$ is a (0,2)-form.

The reduction of the gauge field proceeds as follows:

- $A_{S^2} \to a(y)$: The component A_{S^2} behaves as a scalar with respect to Σ. Its zero mode corresponds to the constant mode on Σ (associated with the generator of $H^0(\Sigma)$). Thus, it survives as a 1-form gauge field $a(y)$ on S^2.
- $A_\Sigma \to X^\mu(y)$: The component A_Σ is a connection 1-form on Σ. In the zero-volume limit, the finite-action principle imposes the flatness condition $F_\Sigma = 0$. The space of flat connections on Σ modulo gauge transformations defines the moduli space $\mathcal{M}_{flat}$. The components of A_Σ thus parametrize the coordinates of this moduli space. Since these components can vary over S^2, they are identified as the scalar fields $X^\mu(y)$ mapping S^2 to the target manifold $\mathcal{M}_{flat}$.

$$A(y) = A_a(y)dx^a + A_i(y)dy^i \longrightarrow \begin{cases} X^\mu(y) \in \Omega^0(S^2, X^*T\mathcal{M}) \\ a(y) \in \Omega^1(S^2) \end{cases} \tag{7.1.3}$$

We also have the shrinking condition on the curvature: $A_\Sigma \Rightarrow F_\Sigma(A_\Sigma) = 0$. The contraction with the vector field v (which is along S^2) gives $\iota_v F_\Sigma = 0$.

2. Fermion Ψ (1-form)

Explicit Algebraic Reduction of $\delta\Psi$

We perform the algebraic reduction of the supersymmetry transformation $\delta\Psi = \iota_v F + id_A\phi$ following the decomposition of fields on $\Sigma \times S^2$.

As above, we decompose the gauge field A and the fermion Ψ into components along Σ and S^2:

$$A = A_\Sigma + a, \tag{7.1.4}$$

$$\Psi = \Psi_\Sigma + \psi. \tag{7.1.5}$$

Here, A_Σ and Ψ_Σ are 1-forms on Σ (which become the scalars X^μ and Ψ^μ in 2D), while a and ψ are 1-forms on S^2.

The curvature 2-form $F = dA + A \wedge A$ decomposes as (Detail see the Appendix):

$$F = F_\Sigma(A_\Sigma) + D_{S^2} A_\Sigma + F(a), \tag{7.1.6}$$

where $D_{S^2} A_\Sigma = d_{S^2} A_\Sigma + [a, A_\Sigma]$ is the mixed component (one leg on Σ, one on S^2). In the limit of shrinking Σ, the finite action principle imposes the flatness condition on Σ:

$$F_\Sigma(A_\Sigma) = 0. \tag{7.1.7}$$

Identifying $A_\Sigma \to X^\mu$, the mixed curvature component becomes the covariant derivative of the scalar:

$$F_{\text{mixed}} = d_{S^2} X^\mu + [a, X^\mu] \equiv DX^\mu. \tag{7.1.8}$$

We compute the variation $\delta\Psi = \iota_v F + i d_A \phi$. We focus on the component along Σ, which corresponds to $\delta\Psi^\mu$. Since the vector field v lies entirely on S^2 ($v^\Sigma = 0$), the contraction ι_v annihilates any pure Σ-form. Thus:

- $\iota_v F_\Sigma = 0$.
- The contraction with the pure S^2 component $F(a)$ yields a 1-form on S^2, do not contributing to $\delta\Psi^\mu$.

The only term contributing to the Σ-component $\delta\Psi^\mu$ is the contraction with the mixed curvature F_{mixed}:

$$(\iota_v F)_\Sigma = \iota_v(F_{\text{mixed}}) = \iota_v(d_{S^2} X^\mu + [a, X^\mu]). \tag{7.1.9}$$

Expanding the contraction:

$$\iota_v(DX^\mu) = v^i \partial_i X^\mu + [v^i a_i, X^\mu]. \tag{7.1.10}$$

This is precisely the definition of the gauge-covariant Lie derivative $\mathcal{L}_v X^\mu$.

For the scalar term $i d_A \phi$, the component along Σ is $D_\Sigma \phi = \partial_\Sigma \phi + i[A_\Sigma, \phi]$. In the reduction limit, $\partial_\Sigma \phi \to 0$. Consistent with the BPS localization locus analysis where $\phi = 0$, this term vanishes.

Thus, we arrive at the reduced transformation law:

$$\boxed{\delta\Psi^\mu = \mathcal{L}_v X^\mu} \tag{7.1.11}$$

3. Fermion χ (2-form)

The 4D 2-form fermion $\chi_{4D} \in \Omega^2(M^4)$ decomposes into three subspaces:

$$\chi_{4D} \in (\Omega^2(\Sigma) \otimes \Omega^0(S^2)) \oplus (\Omega^1(\Sigma) \otimes \Omega^1(S^2)) \oplus (\Omega^0(\Sigma) \otimes \Omega^2(S^2)). \tag{7.1.12}$$

After reduction, we look for 1-form fields on S^2. The only surviving component that matches the required form degree is the mixed term:

$$H^1(\Sigma) \otimes \Omega^1(S^2). \tag{7.1.13}$$

The other components (in $H^2(\Sigma)$ or $H^0(\Sigma)$) do not produce a 1-form on S^2. Thus, the reduction gives:

$$\chi_{4D} \longrightarrow \chi_{2D}(y) \in \Omega^1(S^2, X^* T\mathcal{M}). \tag{7.1.14}$$

Similarly, the 4D bosonic 2-form H reduces to the 2D 1-form boson $H^\mu_{2D}(y)$.

BRST Transformations and 2D Action

We match the 4D BRST algebra to the 2D sigma model algebra.

$$\delta\chi = H \quad \longrightarrow \quad \boxed{\delta\chi^\mu = H^\mu}. \tag{7.1.15}$$

$$\delta H = i\mathcal{L}_v\chi \quad \longrightarrow \quad \boxed{\delta H^\mu = i\mathcal{L}_v\chi^\mu}. \tag{7.1.16}$$

$$\delta A = i\Psi \quad \longrightarrow \quad \boxed{\delta X^\mu = i\Psi^\mu}. \tag{7.1.17}$$

These satisfy the closure relation $\delta^2 = \mathcal{L}_v$.

7.2 Projector Reduction

We now study the constraint $P_\omega^+\,\chi = \chi$ on the mixed part $\chi_{\Sigma S} \in \Omega^1(\Sigma)\otimes\Omega^1(S^2)$. From physics we keep only $\chi_{\Sigma S}$. Since it has one index on Σ and one index on S^2. When Σ becomes very small, the leg on Σ disappears. The leg on S^2 stays. So $\chi_{\Sigma S}$ becomes a one-form on S^2. A one-form on S^2 can be a dynamical fermion in two dimensions.

The field $\chi_{\Sigma\Sigma}$ has two legs on Σ. Both legs disappear after the reduction. It becomes a scalar with no kinetic term. It does not give any physical degrees of freedom.

The field χ_{SS} has two legs on S^2. Such a two-form does not fit the two-dimensional multiplet. It also does not get a kinetic term. So it is not a dynamical field either.

Only the mixed field $\chi_{\Sigma S}$ stays dynamical in the 2D theory. We give an explicit proof later.

Ellipticity means the field can be expanded in eigenmodes on the compact space $M^4 = \Sigma \times S^2$. We reduce the theory by shrinking the internal manifold Σ to zero size. When Σ shrinks, All non-zero modes with respect to Σ acquire large kinetic eigenvalues from the Q-exact term in the localization limit $t \to \infty$. They are exponentially suppressed in the path integral and are therefore integrated out. Only the zero modes on Σ remain dynamical in the 2D theory. Only a finite set of **zero modes** survive to become the fields of the 2D theory. For 1-forms on Σ, these zero modes are the **harmonic 1-forms**. Therefore, we expand the mixed fermion component $\chi_{\Sigma S}$ by using the harmonic basis on Σ and keeping arbitrary field dependence on S^2:

$$\Omega^1(\Sigma) \;\otimes\; \Omega^1(S^2) \;\longrightarrow\; (\text{zero modes on } \Sigma) \;\otimes\; \Omega^1(S^2). \tag{7.2.1}$$

$$\chi_{\Sigma S}(x,y) = \sum_{I=1}^{2g} \psi_I(x)\otimes\eta_I(y) \tag{7.2.2}$$

Here, $\{\psi_I(x)\}$ constitutes an orthonormal basis of harmonic 1-forms on Σ, representing the zero modes. The index I runs over the dimension of the first cohomology group $H^1(\Sigma)$, which is determined by the genus g of the surface as $\dim H^1(\Sigma) = 2g$. The fields $\eta_I(y)$ are the effective dynamical 1-form fields on S^2 in the 2D theory.

To reduce the 4D projector P_ω^+ to 2D, we must analyze how its constituent operators act on this product basis. We start with the 4D constraint equation $P_\omega^+\chi_{\Sigma S} = \chi_{\Sigma S}$. Substituting the explicit form of the 4D projector $P_\omega^+ = \frac{1}{1+h^2}(1 + h *_4 - s^2 m)$, then we get:

$$\frac{1}{1+h^2}\Big[\psi_I\otimes\eta_I + h(J_I{}^J\psi_J)\otimes(*_{S^2}\eta_I) - s^2\psi_I\otimes(m_{S^2}\eta_I)\Big] = \sum_I \psi_I\otimes\eta_I(y) \tag{7.2.3}$$

The reduction relies on decomposing the 4D operators in the projector P_ω^+:

- The 4D Hodge star decomposes as $*_4 = *_\Sigma \otimes *_{S^2}$. On the harmonic basis, $*_\Sigma$ acts as the complex structure tensor $\mathbf{J}$ of the target moduli space (the Jacobian of Σ). In our derivation: $*_\Sigma \psi_I = J_I{}^J \psi_J$.
- The operator m acts trivially on Σ and non-trivially on S^2.

To eliminate the dependence on Σ, we project onto the basis ψ_K using the inner product $\langle \psi_I, \psi_K \rangle_\Sigma = \sum \delta_{KI}$. Taking the inner product with ψ_K on both sides leads to the following vector equation for $\vec{\eta}$. (Detailed algebraic steps are provided in Appendix A):

$$h(*_{S^2} J)\vec{\eta} - s^2 m\vec{\eta} = h^2\vec{\eta} \tag{7.2.4}$$

The result is an effective constraint on the vector of 2D fields $\vec{\eta} = (\eta_1, \ldots, \eta_g)^T$:

$$\boxed{P^+_{2D}\vec{\eta} = \vec{\eta}}, \quad \text{with} \quad P^+_{2D} = \frac{1}{1+h^2}\left(1 + h(*_{S^2}\mathbf{J}) - s^2 m_{S^2}\right). \tag{7.2.5}$$

Note that J acts on the indices of $\vec{\eta}$ while $*_{S^2}$ acts on the differential form component. The term $h(*_{S^2} J)$ couples the spacetime chirality ($*_{S^2}$) with the R-symmetry twist (J). This structure is characteristic of $\mathcal{N} = (2,2)$ supersymmetry. The coefficient $h = \cos\theta$ serves as an interpolating function. At the North Pole ($h = 1$), the projector enforces a specific chirality linked to J; at the South Pole ($h = -1$), this chirality is flipped. This confirms that this operator P^+_{2D} matches the "new self-duality" projector required for the 2D equivariant sigma model.

7.3 Constraint on Other Components ($\chi_{\Sigma\Sigma}, \chi_{SS}$)

We now analyze the reduction of the "unwanted" components of the 4D fermion field χ, decomposing the fields into the subspaces $\Omega^2(\Sigma) \otimes \Omega^0(S^2)$ and $\Omega^0(\Sigma) \otimes \Omega^2(S^2)$. Since the zero modes on Σ are constant or harmonic, we define the expansions as follows using the fixed harmonic basis on Σ:

For the (2,0) component, let ω_Σ be the fixed harmonic 2-form on Σ (the basis) and let $\lambda_0(y)$ be a dynamical scalar field on S^2. The field is expanded as:

$$\chi_{\Sigma\Sigma} = \omega_\Sigma \otimes \lambda_0(y) \tag{7.3.1}$$

For the (0,2) component, the zero mode on Σ is the constant function 1. We let λ_{S^2} denote the fixed 2-form on S^2 (acting as a basis element for the 2-form sector) and let $\omega_0(y)$ be a scalar coefficient field on S^2. The field is expanded as:

$$\chi_{SS} = \underbrace{1}_{\text{basis on } \Sigma} \otimes (\underbrace{\omega_0(y)}_{\text{scalar field}} \cdot \underbrace{\lambda_{S^2}}_{\text{basis on } S^2}) \tag{7.3.2}$$

Note that here ω_0 represents the coefficient field, not a form on Σ.

The action of the operator m and of the 4D Hodge star $*_4$ on these elements is straightforward: since v has no components along Σ, the contraction $\iota_v \omega_\Sigma = 0$, and thus $m\omega_\Sigma = 0$. On S^2, m acts as the identity on the 2-form λ_{S^2}, so $m\lambda_{S^2} = \lambda_{S^2}$; the 4D Hodge star exchanges the volume forms of the two submanifolds, yielding $*_4\omega_\Sigma = \lambda_{S^2}$ and $*_4\lambda_{S^2} = \omega_\Sigma$.

Applying $P^+_\omega(\chi_{\Sigma\Sigma} + \chi_{SS}) = \chi_{\Sigma\Sigma} + \chi_{SS}$ gives:

$$\frac{1}{1+h^2}\left[(\lambda_0 + h\omega_0)\omega_\Sigma + (h\lambda_0 + h^2\omega_0)\lambda_{S^2}\right] = \lambda_0\omega_\Sigma + \omega_0\lambda_{S^2}. \tag{7.3.3}$$

Substituting the explicit form of the projector $P^+_\omega = \frac{1}{1+h^2}(1 + h *_4 - s^2 m)$. Matching the coefficients for the basis ω_Σ and λ_{S^2} independently:

$$\frac{1}{1+h^2}(\lambda_0 + h\omega_0) = \lambda_0,$$
$$\frac{1}{1+h^2}(h\lambda_0 + h^2\omega_0) = \omega_0.$$

The constraint implies that the coefficients λ_0 and ω_0 are not independent. They are locked by the function $h(\theta)$:

$$\omega_0(y) = h(\theta)\lambda_0(y). \tag{7.3.4}$$

Thus, the (2,0) and (0,2) components combine into a single physical mode determined by λ_0:

$$\chi^{(0\oplus 2)} = \lambda_0(\omega_\Sigma + h\lambda_{S^2}). \tag{7.3.5}$$

The Q-exact Term

To perform supersymmetric localization and derive the kinetic terms of the effective 2D theory, we add a Q-exact deformation term to the action. Following standard construction in [12]. The localization functional:

$$S_{loc} = \delta V_{loc} = \delta \int_{M^4} \mathrm{Tr}\left[2(2\Omega - H) \wedge *\chi + (\iota_v F - i d_A(\phi + i(s - \tilde{s})\varphi)) \wedge *\Psi + (\iota_v d_A \varphi + [\phi, \varphi])\eta\right]. \tag{7.3.6}$$

Here, Ω is an auxiliary background field (which we set to zero for simplicity), and the fermions χ, Ψ, η act as Lagrange multipliers enforcing the BPS equations. From the 4D supersymmetry algebra, $\delta\chi = H$ and $\delta H = i\mathcal{L}_v\chi - i[\phi, \chi]$. The kinetic term in the Q-exact action for the locked component $\chi^{(0\oplus 2)} = \lambda_0(\omega_\Sigma + h\lambda_{S^2})$. The relevant term is:

$$S_{kin} = -2i \int_{M^4} \mathrm{Tr}\left[(\mathcal{L}_v \chi^{(0\oplus 2)}) \wedge *\chi^{(0\oplus 2)}\right]. \tag{7.3.7}$$

Substituting the expression for $\chi^{(0\oplus 2)}$ and using the properties $*_4\omega_\Sigma = \lambda_{S^2}$ and $*_4\lambda_{S^2} = \omega_\Sigma$, the integrand simplifies to:

$$(\mathcal{L}_v \chi^{(0\oplus 2)}) \wedge *\chi^{(0\oplus 2)} = (1 + h^2)\lambda_0(v^\mu \partial_\mu \lambda_0)\,\mathrm{vol}_{M^4}. \tag{7.3.8}$$

The action reduces to an integral over S^2:

$$S_{kin} \propto \int_{S^2} (1 + h^2) v^\mu \partial_\mu(\lambda_0^2)\sqrt{g}\, d^2x. \tag{7.3.9}$$

Since v is a Killing vector, $\nabla_\mu v^\mu = 0$. Since $h = \cos\theta$ is invariant along the rotation v, $v^\mu \partial_\mu h = 0$.

$$\implies \nabla_\mu((1 + h^2)v^\mu) = 0. \tag{7.3.10}$$

Therefore, the kinetic integral vanishes identically:

$$\boxed{S_{kin} = 0}. \tag{7.3.11}$$

It serves as a Lagrange multiplier and can be integrated out, imposing constraints on the remaining fields. Consequently, the path integral does not acquire any dynamical contributions from the $\chi_{\Sigma\Sigma}$ and χ_{SS} components, as they lack diagonal kinetic terms. The effective physics of the 2D theory is therefore independent of these "unwanted" sectors and is determined exclusively by the surviving "mixed" component $\chi_{\Sigma S}$ (identified as the physical field χ_{2D}), which constitutes the true dynamical degree of freedom of the sigma model.

7.4 The Non-Q-exact Term

In addition to the Q-exact dynamics, the theory contains a topological observable (Non-Q-exact term) which determines the topological sector and parameters of the 2D theory. We start from Eq. (59) in [12]:

$$S_{non-Q} = \int_{M^4} \mathrm{Tr}\left[hF \wedge F - (\Psi \wedge \Psi + 2\phi F) \wedge \Omega_2 - \phi^2 \Omega_4\right]. \tag{7.4.1}$$

Here $h = \cos\omega(\theta) = \cos\theta$ and Ω_2, Ω_4 are equivariant background forms on S^2. The expansion contains three terms, but the third one vanishes identically due to the geometry of the sphere. As defined in [12] (Eq. 60), the form Ω_4 involves expressions such as $d\kappa \wedge d\kappa$. Since κ is a 1-form on S^2, the form Ω_4 is a 4-form on S^2, and any differential form of degree greater than 2 on a 2-dimensional manifold necessarily vanishes. Thus,

$$\Omega_4 \equiv 0. \tag{5.1.1}$$

Consequently, the term $\varphi_2 \Omega_4$ drops out, and only the first two contributions—the topological term $h\,F \wedge F$ and the coupling term involving Ω—need to be evaluated. For term

$$S_{non-Q} \supset \int_{M^4} \Omega_2 \wedge \Psi \wedge \Psi. \tag{7.4.2}$$

Here Ω_2 is a 2-form on S^2. We decompose $\Psi = \Psi_\Sigma + \Psi_{S^2}$. The term involving the unwanted field Ψ_{S^2} (1-form on S^2) is:

$$\int_{\Sigma \times S^2} \Omega_2 \wedge \Psi_{S^2} \wedge \Psi_{S^2}. \tag{7.4.3}$$

The total form degree on S^2 is 2(from Ω_2)+1(from Ψ_{S^2})+1(from Ψ_{S^2}) = 4. Since the dimension of S^2 is 2, any 4-form vanishes identically:

$$\Omega_2 \wedge \Psi_{S^2} \wedge \Psi_{S^2} = 0. \tag{7.4.4}$$

Thus, the unwanted field Ψ_{S^2} does not contribute to this term.

Reduction of the Surviving Terms

We perform the dimensional reduction on the remaining terms by integrating over Σ.

The Topological Term ($hF \wedge F$)

$$S^{4D}_{top} = \int_{M^4} \cos\theta\, \mathrm{Tr}(F \wedge F). \tag{7.4.5}$$

Using the decomposition $F = F_{\Sigma\Sigma} + F_{\Sigma S} + F_{SS}$, the only component that saturates the volume form (2 legs on Σ, 2 legs on S^2) is $F_{\Sigma S} \wedge F_{\Sigma S}$. Identifying $F_{\Sigma S} \approx dX$, and using the definition of the target space symplectic form $\omega_{\mu\nu} = \int_\Sigma \mathrm{Tr}(\alpha_\mu \wedge \alpha_\nu)$, we get:

$$S^{2D}_{top} = \int_{S^2} \cos\theta\, \omega_{\mu\nu}(X) dX^\mu \wedge dX^\nu. \tag{7.4.6}$$

The Ω_2 Term ($(\Psi^2 + 2\phi F) \wedge \Omega_2$) This term decomposes into a fermionic mass term and a bosonic coupling.

$$S^{4D}_{\Omega_2} = -\int_{M^4} \Omega_2 \wedge \mathrm{Tr}(\Psi \wedge \Psi + 2\phi F). \tag{7.4.7}$$

- **Fermionic part ($\Psi \wedge \Psi$):** We decompose $\Psi = \Psi_\Sigma + \Psi_{S^2}$. As proven in the decoupling section, $\Psi_{S^2} \wedge \Psi_{S^2} \wedge \Omega_2 = 0$ due to degree reasons (4-form on S^2). The only surviving term involves Ψ_Σ:

$$\int_{M^4} \Omega_2 \wedge \Psi_\Sigma \wedge \Psi_\Sigma = \int_{S^2} \Omega_2 \left(\int_\Sigma \mathrm{Tr}(\Psi_\Sigma \wedge \Psi_\Sigma) \right). \tag{7.4.8}$$

  Identifying $\int_\Sigma \Psi^2_\Sigma$ with the pullback of the symplectic form, this yields the 2D mass term:

$$S^{2D}_{mass} = -\int_{S^2} \Omega_2\, \omega_{\mu\nu} \Psi^\mu \Psi^\nu. \tag{7.4.9}$$

- **Bosonic part ($2\phi F$):** Similarly, only the component $F_{\Sigma\Sigma}$ (2 legs on Σ) contributes:

$$S^{2D}_{\phi F} = -\int_{S^2} 2\phi\, \Omega_2 \left(\int_\Sigma \mathrm{Tr}(F_{\Sigma\Sigma}) \right). \tag{7.4.10}$$

  This term couples the scalar ϕ to the magnetic flux on Σ, which corresponds to the moment map $\mu(X)$ (or D-term) of the target space. On the moduli space of flat connections ($\mathcal{M}_{flat}$), $F_{\Sigma\Sigma} \approx 0$, so this term vanishes in that specific limit, but it is part of the general equivariant action.

7.5 The Q-exact Term

We explicitly derive the 2D kinetic terms for the fermions from the Q-exact action.

Starting from the Q-exact term involving the mixed component $\chi_{\Sigma S}$ and the auxiliary field H:

$$S_{\chi H} = \delta \int_{S^2} \mathrm{Tr}\left(-2\chi_{\Sigma S} \wedge *H\right). \tag{7.5.1}$$

Using the BRST transformations $\delta\chi = H$ and $\delta H = i\mathcal{L}_v \chi$, the variation yields a bosonic $|H|^2$ term and a fermionic kinetic term. Focusing on the fermionic part:

$$S^{kin}_\chi = -2 \int_{S^2} \mathrm{Tr}\left(\chi_{\Sigma S} \wedge *(i\mathcal{L}_v \chi)\right). \tag{7.5.2}$$

This provides the standard kinetic term for the 1-form fermion χ_{2D}.

Starting from the Q-exact term involving Ψ:

$$S_\Psi = \delta \int_{M^4} \mathrm{Tr}\left[(\iota_v F - i d_A \phi) \wedge *\Psi\right]. \tag{7.5.3}$$

The key step is to evaluate the variation of the bosonic factor. Using the relations $\delta A = i\Psi$ and $\delta\phi = \iota_v \Psi$, and applying Cartan's formula $\mathcal{L}^A_v = d_A \iota_v + \iota_v d_A$, we find:

$$\delta(\iota_v F - i d_A \phi) = i\mathcal{L}^A_v \Psi. \tag{7.5.4}$$

Substituting this back and integrating over Σ (which produces the metric factor $G_{\mu\nu}$), we obtain the kinetic term for the scalar fermion Ψ^μ:

$$S^{kin}_\Psi = i \int_{S^2} \sqrt{g} G_{\mu\nu} \Psi^\mu \wedge *(\mathcal{L}_v \Psi^\nu). \tag{7.5.5}$$

7.6 The Effective 2D Action

Based on the dimensional reduction from 4D on $\Sigma \times S^2$ (shrinking Σ), the total 2D effective action S_{2D} is decomposed into a bosonic part and a fermionic part:

$$S_{2D} = S_{bos} + S_{ferm}. \tag{7.6.1}$$

Combining all the derived components (S_{bos} and S_{ferm}), the total effective action is:

$$\begin{aligned} S_{2D} = \int_{S^2} \Big[& -\sqrt{g} G_{\mu\nu} (P^+ dX^\mu) \wedge *(P^+ dX^\nu) + \cos\theta\, \omega_{\mu\nu} dX^\mu \wedge dX^\nu \\ & - i2\chi_{\Sigma S} \wedge *\mathcal{L}_v \chi + i\sqrt{g} G_{\mu\nu} \Psi^\mu \wedge *(\mathcal{L}_v \Psi^\nu) - \Omega_2 \omega_{\mu\nu} \Psi^\mu \Psi^\nu \Big]. \end{aligned} \tag{7.6.2}$$

This matches the structure of the equivariant sigma model action. We have rigorously proven that the dimensional reduction of the 4D $\mathcal{N} = 2$ supersymmetric Yang-Mills theory on $\Sigma \times S^2$ defined by the flipping projector P^+_ω precisely yields the 2D $\mathcal{N} = (2, 2)$ equivariant sigma model on S^2, demonstrating that the 2D supersymmetry structure and action are exact descendants of the 4D theory upon the decoupling of non-dynamical fields.

Conclusion

We have shown how a four-dimensional twisted gauge theory on a product manifold can be reduced to an effective two-dimensional theory. Although the resulting models are not strictly topological, they fit naturally into an equivariant framework, where the scalar supercharge closes on a geometric symmetry.

The main significance of this work lies in providing a clear higher-dimensional origin for known two-dimensional theories. This perspective explains the special geometric structures that appear in the two-dimensional action and places them in a unified framework together with other equivariant constructions.

A Operator Decomposition

Using the product metric on $M^4 = \Sigma \times S^2$, the operators act on the basis $\psi_I \otimes \eta_a$ as follows:

- **Hodge Star ($*_4$):** $*_4 = *_\Sigma \otimes *_{S^2}$. The action of $*_\Sigma$ on the harmonic basis corresponds to the complex structure matrix $(\mathbf{J})_I^J$:

$$*_\Sigma \psi_I = \sum_J (\mathbf{J})_I^J \psi_J. \tag{A.0.1}$$

  Thus, the action on the tensor product is:

$$*_4(\psi_I \otimes \eta_I) = \sum_J (\mathbf{J})_I^J \psi_J \otimes (*_{S^2}\eta_I). \tag{A.0.2}$$

- **Operator m:** Since v lies on S^2, m acts as identity on Σ:

$$m(\psi_I \otimes \eta_I) = \psi_I \otimes (m_{S^2}\eta_I). \tag{A.0.3}$$

Reduction via Inner Product Substituting these into the 4D equation $P_\omega^+ \chi = \chi$:

$$\frac{1}{1+h^2} \sum_I \left[\psi_I \otimes \eta_I + h \sum_J (\mathbf{J})_I^J \psi_J \otimes (*_{S^2}\eta_I) - s^2 \psi_I \otimes (m_{S^2}\eta_I) \right] = \sum_I \psi_I \otimes \eta_I. \tag{A.0.4}$$

To reduce to 2D, we take the inner product $\langle \psi_K, \cdot \rangle_\Sigma = \int_\Sigma \psi_K \wedge *_\Sigma(\cdot)$ on both sides. Using the orthonormality $\langle \psi_K, \psi_I \rangle = \delta_{KI}$, the sums collapse:

$$\frac{1}{1+h^2} \left[\eta_K + h \sum_I (\mathbf{J})_K^I (*_{S^2}\eta_I) - s^2 (m_{S^2}\eta_K) \right] = \eta_K. \tag{A.0.5}$$

Rearranging terms to group the fields:

$$h \sum_I (\mathbf{J})_K^I (*_{S^2}\eta_I) - s^2 (m_{S^2}\eta_K) = h^2 \eta_K. \tag{A.0.6}$$

Vector Notation Defining $\vec{\eta}$ as the column vector of 1-forms, we write the system as:

$$[h(*_{S^2}\mathbf{J}) - s^2 m_{S^2}]\vec{\eta} = h^2 \vec{\eta}. \tag{A.0.7}$$

Adding $\vec{\eta}$ to both sides and dividing by $(1+h^2)$ yields the final projector equation:

$$\frac{1}{1+h^2} \left[1 + h(*_{S^2}\mathbf{J}) - s^2 m_{S^2} \right] \vec{\eta} = \vec{\eta}. \tag{A.0.8}$$

Note that the sign of the $h(*_{S^2}\mathbf{J})$ term depends on the specific convention of the complex structure action on the dual space, matching the result P_{2D}^+ in the main text.

B Detailed Algebraic Derivation of Dimensional Reduction

In this appendix, we provide a comprehensive algebraic derivation of the effective 2D supersymmetry transformation laws starting from the 4D twisted theory. We explicitly perform the dimensional reduction on the product manifold $M^4 = \Sigma \times S^2$ in the limit where the volume of Σ vanishes.

Geometric Setup and Field Decomposition

Let $M^4 = \Sigma \times S^2$. We introduce local coordinates $x^\mu = (x^a, y^i)$, where x^a $(a = 1, 2)$ are coordinates on Σ and y^i $(i = 1, 2)$ are coordinates on S^2. The exterior derivative decomposes as $d = d_\Sigma + d_{S^2}$, where $d_\Sigma = dx^a \partial_a$ and $d_{S^2} = dy^i \partial_i$.

The Killing vector field v generating the rotational isometry lies entirely on S^2. Thus, its components satisfy:

$$v = v^i(y) \frac{\partial}{\partial y^i} \implies v^a = 0, \quad v^i \neq 0. \tag{B.0.1}$$

We decompose the 4D Lie-algebra valued gauge field A and the fermion Ψ based on their form degree on Σ:

$$A(x, y) = \underbrace{A_a(x, y) dx^a}_{A_\Sigma} + \underbrace{A_i(x, y) dy^i}_{a}, \tag{B.0.2}$$

$$\Psi(x, y) = \underbrace{\Psi_a(x, y) dx^a}_{\Psi_\Sigma} + \underbrace{\Psi_i(x, y) dy^i}_{\psi}. \tag{B.0.3}$$

In the dimensional reduction limit ($\mathrm{vol}(\Sigma) \to 0$), we retain only the zero modes along Σ. This imposes two conditions:

1. **Independence**: All fields are independent of Σ, i.e., $\partial_a \Phi = 0$ for any field Φ.
2. **Flatness**: The curvature along Σ vanishes to satisfy the finite action principle, i.e., $F_{ab} = 0$.

Under this reduction, we identify the components with the 2D effective fields:

- $A_\Sigma \to X(y)$ (Scalar field on S^2, valued in the adjoint bundle).
- $A_{S^2} \to a(y)$ (Gauge field on S^2).
- $\Psi_\Sigma \to \Psi^\mu(y)$ (Scalar fermion on S^2).

Decomposition of the Curvature Tensor

The 4D curvature 2-form is defined by $F = dA + A \wedge A$. We expand this in terms of the decomposed fields:

$$F = (d_\Sigma + d_{S^2})(A_\Sigma + a) + (A_\Sigma + a) \wedge (A_\Sigma + a). \tag{B.0.4}$$

Expanding the terms linearly and quadratically:

$$F = \underbrace{(d_\Sigma A_\Sigma + A_\Sigma \wedge A_\Sigma)}_{\text{Type (2,0)}} + \underbrace{(d_{S^2} A_\Sigma + d_\Sigma a + A_\Sigma \wedge a + a \wedge A_\Sigma)}_{\text{Type (1,1)}} + \underbrace{(d_{S^2} a + a \wedge a)}_{\text{Type (0,2)}}. \tag{B.0.5}$$

Using the reduction condition $\partial_a(\dots) = 0$, the term $d_\Sigma a = \partial_a a_i dx^a \wedge dy^i$ vanishes. The commutator is $[a, A_\Sigma] = a \wedge A_\Sigma + A_\Sigma \wedge a$. Thus, the curvature decomposes into three distinct sectors:

$$F = F_{\Sigma\Sigma} + F_{\text{mixed}} + F_{SS}, \tag{B.0.6}$$

where:

- $F_{\Sigma\Sigma} = F_\Sigma(A_\Sigma)$ is the curvature on Σ.
- $F_{SS} = F(a)$ is the curvature on S^2.

- F_{mixed} is the term with one leg on Σ and one on S^2:

$$F_{\text{mixed}} = d_{S^2} A_\Sigma + [a, A_\Sigma]. \tag{B.0.7}$$

Identifying A_Σ with the scalar X, the expression $d_{S^2} X + [a, X]$ is precisely the gauge-covariant derivative on S^2:

$$F_{\text{mixed}} = DX, \quad \text{where } D = d_{S^2} + [a, \cdot]. \tag{B.0.8}$$

Derivation of the Transformation Law $\delta\Psi^\mu$

We start with the 4D supersymmetry variation for the fermion Ψ:

$$\delta\Psi = \iota_v F + i d_A \phi. \tag{B.0.9}$$

We aim to find the transformation for the scalar fermion Ψ^μ, which corresponds to the component of $\delta\Psi$ along Σ. In local coordinates, this is the a-component $(\delta\Psi)_a$.

The Contraction Term $\iota_v F$

The contraction operator ι_v acts on the 2-form $F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$. The component along the x^a direction is:

$$(\iota_v F)_a = v^\mu F_{\mu a} = v^b F_{ba} + v^i F_{ia}. \tag{B.0.10}$$

Since the vector field v has no components along Σ ($v^b = 0$), the first term vanishes identically. We are left with the contraction of the mixed curvature component:

$$(\iota_v F)_a = v^i F_{ia}. \tag{B.0.11}$$

From Eq. (B.0.8), the mixed curvature in components is $F_{ia} = D_i X = \partial_i X + [a_i, X]$. Substituting this back:

$$(\iota_v F)_a = v^i (D_i X). \tag{B.0.12}$$

Geometrically, the contraction of a vector field with the covariant derivative defines the **covariant Lie derivative** $\mathcal{L}_v$:

$$\mathcal{L}_v X := v^i D_i X = v^i \partial_i X + [v^i a_i, X]. \tag{B.0.13}$$

Thus, we obtain the relation:

$$(\iota_v F)_\Sigma = \mathcal{L}_v X. \tag{B.0.14}$$

The Scalar Gradient Term $i d_A \phi$

The covariant derivative of the scalar field ϕ is $d_A \phi = d\phi - i[A, \phi]$. We consider its component along Σ:

$$(d_A \phi)_a = \partial_a \phi - i[A_a, \phi]. \tag{B.0.15}$$

In the dimensional reduction limit:

- The derivative term vanishes: $\partial_a \phi = 0$.
- The gauge field component becomes the scalar: $A_a \to X$.

Thus:

$$i(d_A\phi)_a = i(-i[X,\phi]) = [X,\phi]. \tag{B.0.16}$$

Combining the results, the effective transformation for the scalar fermion is:

$$\delta\Psi^\mu = \mathcal{L}_v X + [X,\phi]. \tag{B.0.17}$$

Consistent with the BPS localization locus analysis where the scalar ϕ is set to zero (or absorbed into the definition of the complexified connection), we recover the simplified equivariant transformation:

$$\boxed{\delta\Psi^\mu = \mathcal{L}_v X^\mu} \tag{B.0.18}$$

This concludes the algebraic derivation of the 2D effective supersymmetry variations from the 4D parent theory.

C Some Necessary Mathematical Knowledge

C.1 Connections and Curvature

A connection on a principal G-bundle is a $\mathfrak{g}$-valued 1-form $A \in \Omega^1(P,\mathfrak{g})$ satisfying the following conditions:

1. **Fundamental property**: For any fundamental vector field $X^\#$ corresponding to an element $X \in \mathfrak{g}$, we have:

   $$A(X^\#) = X$$

   where $X^\#$ is the fundamental vector field associated with the Lie algebra element X, and A is a 1-form on the total space P of the principal bundle.

2. **Transformation under gauge group action**: The connection 1-form transforms under the action of the gauge group G as:

   $$R_g^* A = \mathrm{Ad}(g^{-1})A$$

   where R_g is the right action of $g \in G$ on the principal bundle, and $\mathrm{Ad}(g^{-1})$ denotes the adjoint action of the Lie group G on its Lie algebra $\mathfrak{g}$.

3. **Curvature**: The curvature F_A of the connection is a $\mathfrak{g}$-valued 2-form defined by:

   $$F_A = dA + A \wedge A$$

   The curvature F_A satisfies the transformation rule:

   $$R_g^* F_A = \mathrm{Ad}(g^{-1})F_A$$

   This shows how the curvature transforms under a gauge transformation.

4. **Bianchi identity**: The curvature F_A satisfies the **Bianchi identity**:

   $$d_A F_A = dF_A + [A \wedge F_A] = 0$$

   where d_A denotes the covariant derivative associated with the connection. This identity expresses the integrability condition for the curvature.

5. **Covariant derivative**: The covariant derivative acting on $\mathfrak{g}$-valued forms ω is defined as:
$$d_A\omega = d\omega + [A \wedge \omega]$$
This is the generalization of the derivative for forms in the presence of a connection.

   Let $P \to M$ be a principal G-bundle over a compact manifold M. The space of all connections A is an affine space modeled on $\Omega^1(M, \mathrm{ad}P)$. The gauge group g acts on A by:
$$g \cdot A = g^{-1}Ag + g^{-1}dg$$
The moduli space of interest is the quotient:
$$\mathcal{M} = \{A \in \mathcal{A} : F_A^+ = 0\}/\mathcal{G}$$
where F_A^+ is the self-dual part of the curvature.

   For an ASD connection on a 4-manifold, the virtual dimension of the moduli space is given by:
$$\dim\mathcal{M} = 8p_1(E) - 3(\chi(M) + \sigma(M))$$
where $\chi(M)$ is the Euler characteristic and $\sigma(M)$ is the signature.

C.2 Donaldson Invariants

For a smooth, compact 4-manifold M, the Donaldson invariants are maps
$$\gamma : H_2(M) \times \cdots \times H_2(M) \to \mathbb{Q}$$
defined by integration over the moduli space of anti-self-dual (ASD) connections $\mathcal{M}$:
$$\gamma(\Sigma_1, \ldots, \Sigma_d) = \int_{\mathcal{M}} \mu(\Sigma_1) \wedge \cdots \wedge \mu(\Sigma_d),$$
where $\mu : H_2(M) \to H^2(\mathcal{M})$ is the so-called μ-map, which associates to a 2-cycle $\Sigma \subset M$ a cohomology class on the moduli space $\mathcal{M}$.

From the physical perspective, in Donaldson-Witten theory, the correlation functions of certain observables $O(\Sigma_i)$ are computed via a path integral:
$$\langle O(\Sigma_1) \cdots O(\Sigma_d) \rangle = \int \mathcal{D}A\, e^{-S_{\mathrm{TQFT}}[A]}\, O(\Sigma_1) \cdots O(\Sigma_d),$$
where the action S_{TQFT} is Q-exact due to the topological twisting of the $N = 2$ supersymmetric Yang-Mills theory, and the path integral localizes to the moduli space $\mathcal{M}$ of ASD connections.

Mathematically, this can be understood as an infinite-dimensional generalization of the Euler class. In finite dimensions, the Euler number of a vector bundle $E \to M$ can be represented as an integral of the Mathai–Quillen Thom form:
$$\int_M e(E) = \int_E \Phi,$$
where Φ is a Gaussian-like differential form depending on the curvature of E. In the infinite-dimensional setting of Donaldson-Witten theory, the space of all connections modulo gauge transformations $\mathcal{A}/\mathcal{G}$ plays the role of a "vector bundle", and the moduli space $\mathcal{M}$ corresponds to the zero locus of a section (the ASD equation). Donaldson invariants are in some sense an infinite dimensional Euler number, and that the path integral representation is an infinite dimensional version of the Mathai-Quillen formula.

C.3 Topological Winding Number in Gauge Theories

In four-dimensional Euclidean spacetime, finite-action gauge field configurations in non-Abelian gauge theories are classified by a topological invariant known as the winding number (or Pontryagin index) $v \in \mathbb{Z}$. This integer characterizes the homotopy class of the gauge field configuration at spatial infinity and is robust against any continuous deformation that preserves the finite-action condition.

Two equivalent integral expressions for the winding number exist. The first is a surface integral over the 3-sphere S^3 at infinity [9]:

$$v = \frac{1}{24\pi^2} \int_{S^3} d^3 S_\mu \, \epsilon^{\mu\nu\lambda\sigma} \mathrm{Tr} \left[(A_\nu \partial_\lambda A_\sigma + \frac{2}{3} i A_\nu A_\lambda A_\sigma) \right], \tag{C.3.1}$$

which measures the mapping from the spatial boundary S^3 into the gauge group manifold. The second is a gauge-invariant volume integral over all spacetime:

$$v = \frac{1}{32\pi^2} \int d^4x \ \mathrm{Tr}[F_{\mu\nu} \tilde{F}^{\mu\nu}], \tag{C.3.2}$$

where $F_{\mu\nu}$ is the non-Abelian field strength tensor and $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ is its Hodge dual.

This structure is inherently non-Abelian. For an Abelian gauge theory like U(1), all mappings from S^3 into the group are topologically trivial ($v \equiv 0$), as the group manifold S^1 does not support such non-trivial winding. In contrast, for a simple non-Abelian group G (e.g., SU(N) for $N \geq 2$), Bott's theorem ensures that the topology is governed by its SU(2) subgroups, making the winding number a universal feature. Consequently, the gauge group topology implies a vacuum structure comprised of disconnected sectors, labeled by v.

C.4 Algebraic Topology and Homology Theory

Cohomology Theories For a smooth manifold M, de Rham cohomology is defined as:

$$H^k_{\mathrm{dR}}(M) = \left\{ \omega \in \Omega^k(M) : d\omega = 0 \right\} / \left\{ d\eta : \eta \in \Omega^{k-1}(M) \right\}$$

The cup product $\smile$: $H^k(M) \times H^l(M) \to H^{k+l}(M)$ makes cohomology a graded ring. For compact oriented n-manifolds, Poincaré duality gives isomorphisms:

$$H^k(M) \cong H^{n-k}(M)$$

where $H^k(M)$ is the k-dimensional cohomology group of the manifold M, and $H^{n-k}(M)$ is the $(n-k)$-dimensional homology group of the manifold M, assuming the manifold M has dimension n.

C.5 Topological Twist

A topological twist modifies the Lorentz representations of the fields using an embedding of the Lorentz group into the R-symmetry group.

$$SO(4) \times SU(2)_R \to SU(2)_+ \times SU(2)_- \times SU(2)_R \to SU(2)_+ \times SU(2)_{\mathrm{diag}}$$

The supercharge becomes a scalar, and the theory becomes topological, which the scalar supercharge Q satisfying

$$Q^2 = 0,$$

either strictly or modulo gauge transformations. After twisting, the fields decompose into forms of different degrees:

$$\mathcal{F}^\bullet = \bigoplus_{p=0}^{n} \mathcal{F}^{(p)}$$

the Q acts as a cohomological differential:

$$Q : \mathcal{F}^{(p)} \to \mathcal{F}^{(p+1)}, \quad Q^2 = 0.$$

and the path integral localizes to the moduli space of instantons:

$$Z = \int_M e^{-S_{\text{eff}}} \frac{1}{\sqrt{\det D}} \prod_i O_i$$

where D is the deformation operator and S_{eff} is the effective action on M.

C.6 Path Integral and Atiyah-Bott Localization

The classical action is an "integral" over a finite dimensional manifold, the quantum path integral is an "integral" over a space of fields on that manifold, which typically is infinite dimensional. The formal path integral measure is constructed as:

$$\mathcal{D}\phi = \prod_x d\phi(x) \sqrt{\det G(\phi(x))} \tag{C.6.1}$$

where G is the metric on field space. For gauge theories, one must divide by the volume of the gauge group:

$$Z = \int \frac{\mathcal{D}A}{\text{Vol}(\mathcal{G})} e^{-S[A]} \tag{C.6.2}$$

Atiyah-Bott Localization

For a Q-exact action $S = \{Q, V\}$ with Q-invariant measure, the path integral localizes to:

$$Z = \sum_{\text{fixed components}} \int_{\mathcal{M}_i} \frac{e^{-S_{\text{eff}}}}{\text{Euler}(\mathcal{N}_i)} \sqrt{\det D_i} \tag{C.6.3}$$

where $\mathcal{M}_i$ denotes the moduli space associated with the i-th fixed component, $\mathcal{N}_i$ is the normal bundle to the i-th fixed component, and D_i is the quadratic fluctuation operator.

C.7 Atiyah–Singer Index Theorem

In Witten-type topological quantum field theories, many physical quantities are computed as integrals over moduli spaces of solutions to certain differential equations (e.g., instanton equations, generalized self-dual equations). These moduli spaces are often formally defined as the zero locus of a differential operator

$$D : \Gamma(E_1) \longrightarrow \Gamma(E_2),$$

where E_1 and E_2 are vector bundles over the manifold M. The formal dimension of the moduli space, also called the *virtual dimension*, is given by the index of D:

$$\text{vdim}(\mathcal{M}) = \text{index}(D) := \dim \ker D - \dim \text{coker} D.$$

The Atiyah–Singer index theorem provides a topological formula for the analytical index of an elliptic differential operator D on a compact manifold M:

$$\operatorname{index}(D) = \int_M \operatorname{ch}(E_1) \wedge \operatorname{Td}(TM) - \int_M \operatorname{ch}(E_2) \wedge \operatorname{Td}(TM),$$

or, more generally, for a single elliptic complex

$$0 \longrightarrow E_0 \xrightarrow{D_0} E_1 \xrightarrow{D_1} \cdots \xrightarrow{D_{n-1}} E_n \longrightarrow 0,$$

the index is

$$\operatorname{index}(D) = \sum_{i=0}^{n} (-1)^i \dim H^i(D),$$

where $H^i(D)$ denotes the cohomology at step i of the elliptic complex.

References

[1] M. Atiyah. Topological quantum field theory. *Publications Mathématiques de l'IHÉS*, 68:175–186, 1988. doi:10.1007/BF02698547.

[2] A. Schwarz. The partition function of degenerate quadratic functional and Ray–Singer invariants. *Letters in Mathematical Physics*, 2(3):247–252, 1978. 10.1007/BF00406412.

[3] S. K. Donaldson, *Polynomial Invariants for Smooth Four-Manifolds*, Topology **29** (1990), no. 3, 257–315, doi.org/10.4310/jdg/1214437665.

[4] E. Witten. Topological quantum field theory. *Communications in Mathematical Physics*, 117:353–386, 1988. doi:10.1007/BF01223371.

[5] K.Hori et al. "Mirror Symmetry." *Clay Mathematics Monographs, Vol. 1, AMS*, 2003. Available at: `http://www.claymath.org/publications/mirror-symmetry`.

[6] R. Dijkgraaf. E. Witten Topological gauge theories and group cohomology. *Communications in Mathematical Physics*, 129, 393–429, 1990. doi:10.1007/BF02096988

[7] D. Grabovsky. Chern–Simons Theory in a Knotshell. 2022. Available at: `https://web.physics.ucsb.edu/~davidgrabovsky/files-notes/CS%20and%20Knots.pdf`,

[8] S. P. Martin, *A Supersymmetry Primer*, arXiv:hep-ph/9709356 [hep-ph]

[9] S. Coleman. Aspects of Symmetry: Selected Erice Lectures. *Cambridge University Press*, 1985. Available at:

`https://www.cambridge.org/core/books/aspects-of-symmetry/7791D1FCFD0FA0CD1CC54C961E2DF470`

[10] M. Bershadsky, A. Johansen, V. Sadov and C. Vafa, *Topological Reduction of 4D SYM to 2D σ-Models*, arXiv:hep-th/9501096.

[11] V. Pestun, *Localization of gauge theory on a four-sphere*, Commun. Math. Phys. **313**, 71-129 (2012), arXiv:0712.2824 [hep-th].

[12] G. Festuccia, J. Qiu, J. Winding and M. Zabzine, *Twisting with a Flip (the Art of Pestunization)*, arXiv:1812.06473 [hep-th].

[13] V. Alekseev, G. Festuccia, V. Mishnyakov, N. Terziev and M. Zabzine, *Localizing non-linear $\mathcal{N} = (2, 2)$ sigma model on S^2*, arXiv:2202.10501 [hep-th].